Intensions and Plurality*

Ido Benbaji-Elhadad
University of Tübingen

Abstract Schmitt (2023) recently argued that no natural language expressions denote pluralities of possible worlds, based on the observation that expressions previously argued to do so do not give rise to cumulative inferences, which diagnose plurality. This paper reveals an intimate connection between Schmitt's conclusion and another assumption that seems entirely independent: The commonly accepted claim that a DP cannot take wide quantificational scope over an intensional operator while its restrictor is evaluated opaquely within that operator's domain of quantification. This assumption is justified by examining scenarios where such a scopal configuration would render a given intensional sentence true, yet that sentence intuitively seems false. I demonstrate that without Schmitt's ban on cumulativity with worlds, standard cumulation mechanisms applied to certain intensional sentences generate truth-conditions that are verified by exactly those scenarios in which these sentences are judged to be false, and which are standardly used to illustrate the absence of wide-scope opaque scopal configurations. The argument not only provides independent support for Schmitt's conclusion but generalizes it: While Schmitt shows that particular expressions previously argued to denote world pluralities should not be so analyzed, this paper illustrates that any system which makes intensions available for cumulation suffers from a problem similar in nature to the one Schmitt observes. I conclude by speculating whether the ban on cumulation with worlds might follow from independent principles, though a comprehensive defense of this approach is left for future work.

1 Introduction

At the heart of this paper is a tension between two assumptions that are quite common in the literature. First, that predicates of all kinds can be true of pluralities in virtue of being true of their atoms. For instance, in a scenario in which Jane arrived and Mary arrived, one can truthfully say that *Mary and Jane arrived*. Similarly, if it is the case that Jane fed Fido and that Mary fed Bob, then the sentence *Jane and Mary fed Fido and Bob* seems true. Thus, our semantic theory must posit a mechanism capable of accommodating such inferences from truths about individuals to truth about pluralities thereof, often called *cumulative inferences*. The second assumption is that intensional operators, like modal adverbs and attitude predicates,

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denote functions that take as their input *intensions*; namely, functions on the domain of possible worlds. Thus, our semantic theory must posit a mechanism capable of introducing intensions into a semantic derivation when so required by the lexical items employed in that derivation.

Both of these assumptions are often treated as truisms, or, at the very least as rather unobjectionable in and of themselves. Recently, however, Schmitt (2023) observed an interaction that is quite objectionable: In particular, she shows that expressions previously argued to denote *pluralities* of possible worlds seem unable to partake in cumulative relations. Schmitt concludes that no natural language expressions denote world pluralities; a conclusion that contradicts a popular, and quite entrenched position in the literature, which attempts to derive certain inferences of intensional sentences from the assumption that certain linguistic expressions in these sentences denote pluralities of exactly this sort (e.g., Schlenker 2004; Križ 2019; Agha & Jeretič 2022).

This paper argues that Schmitt's conclusion interacts in a surprising way with another theoretical claim that at first blush seems wholly unrelated to the nature of that conclusion. This is the claim that DPs cannot take wide quantificational scope relative to an intensional operator if their restrictor is evaluated *opaquely*, in the intensional environment created by the operator. The claim is old, quite broadly accepted—see, for instance, Ioup 1975; Keshet 2008, 2010; von Fintel & Heim 2011; Keshet & Schwarz 2019; Benbaji 2021; Benbaji-Elhadad 2026b for arguments in favor of it, and Demirok 2019; Elliott 2023; Mirrazi 2024 for recent works that take it for granted—and is regularly justified by pointing out that certain sentences seem *false* in scenarios which, had wide quantificational scope and opaque evaluation gone hand in hand, would render these sentences *true* on a wide scope opaque construal.[1]

I show that this widely-accepted generalization is only predicted to hold if one accepts Schmitt's (2023) surprising conclusion about the absence of world pluralities: If the mechanism we posit to derive cumulative inferences is allowed to apply to intensions in the course of the semantic derivation, this will render intensions true of pluralities of *worlds* by virtue of being true of the atomic members of these pluralities. Certain cumulative inferences with pluralities of worlds are in turn *entailed* by those readings that would be derived from the scopal configuration generally assumed not to exist. And, of course, any scenarios that verifies those latter readings also verify anything they entail. Thus, without a ban on cumulative readings with

1 As is often the case, of course, the position is not *universally* accepted: Fodor (1970) famously argued that DPs can indeed scope above an intensional operator that determine the evaluation domain of their restrictors; a position that has been recently revived by Szabó (2010, 2011). The common view is defended against Szabó's counter-examples in Benbaji 2021; Benbaji-Elhadad 2026b, and I set aside these cases for the purpose of the current paper.

worlds, we wrongly predict certain sentences to be judged true, in scenarios that as a matter of fact render them false.

More specifically, I will consider a sentence like (1), uttered in a scenario in which only one victim is trapped under the rubble, but two particular individuals could each be that victim. Imagine, for instance, that a building collapsed with just one security guard inside it, but that rescue forces are unsure which of two possible guards was on duty during the collapse. The example is judged to be *false* in such a scenario. This lends support to the teachings of orthodoxy, of course, according to which the DP "*two victims*" cannot have its numeral "*two*" scope above the modal "*might*" while its adjacent NP "*victims*" is evaluated in the epistemically accessible worlds quantified over by the modal. If it could have had that scopal configuration, we would expect the sentence to have a reading paraphrasable as in (2), which is *true* in the above scenario.

(1) Two victims might be trapped under the rubble.

(2) Two individuals (each) might be a victim trapped under the rubble.

I will show that on the common assumption that particular constituents in the syntactic structure assigned to (1) denote intensions, we obtain a reading for the sentence that is entailed by (2), once these intensions interact with standard mechanisms that derive cumulative inferences.

Here is a fairly non-technical preview of the argument: Cumulation mechanisms derive a true reading for a sentence like "*Two girls fed two cats*", even if there are no two girls who each fed two cats, as long as some girl fed some cat, and some other girl fed some other cat. More generally, these mechanisms allow a relation (like the one denoted by "*fed*" above) to hold between two plural indefinite quantifiers, if it holds between the members of the pluralities witnessing each indefinite in a way that no member of either witness is "left out" (Champollion 2020).

If the predicate "*victim*" in (1) itself denotes a relation, true of an individual and a world if that individual is a victim in that world, cumulation should render that relation true of two *pluralities*, of individuals and worlds, just in case: (i) for each part of the individual plurality there is a world in the world plurality, in which it is a victim, and (ii) for each world in the world plurality there is a part of the individual plurality that is a victim in it. Then, as long as we allow the modal "*might*" to quantify existentially over pluralities of worlds, (1) should be rendered true by the existence of one world in which some individual is a victim, and another world in which some other individual is a victim. But this is precisely what goes on in the collapse scenario discussed above. Thus, (1) is predicted to have a reading that is equivalent to what would be derived by positing wide scope opaque scopal configurations, contrary to fact. To block the unattested reading, then, we must block the interaction between intensions and cumulation mechanisms.

The argument in this paper thus provides a new motivation for Schmitt's (2023) conclusion: If we want to capture the empirical landscape *vis-a-vis* intensionality and scope correctly, we need to take for granted the conclusion that Schmitt motivates independently. However, the argument also goes a step further. Schmitt's argument examines *particular* expressions previously claimed to denote world pluralities—e.g., *if*-clauses (Schlenker 2004), doxastic attitudes (Križ 2019)—and shows that the world plurality hypothesis overgenerates unattested cumulative readings of sentences containing these expressions. But Schmitt also notes that proponents of the world plurality hypothesis might attribute the absence of such cumulative readings to a stipulated quirk in the syntax or lexical semantics of the relevant expressions. While this response is conceptually problematic—it renders world pluralities effectively undiagnoseable by appealing to cumulativity (and, perhaps, the world-plurality hypothesis unfalsifiable)—it has not yet been shown to be empirically inadequate.

I show that, indeed, the problem Schmitt identifies cannot be handled in this way, as it does not only concern those particular modal expressions. Rather, we can apply the logic of Schmitt's argument to illustrate that the mere presence of elements that denote intensions in the structural representation of a sentence, when paired with the pluralizing mechanisms that derive cumulative inferences, leads to readings long thought to be unattested.

After presenting the main argument spelled-out above, in the more speculative part of this paper, I will explore whether we can block unattested cumulative inferences with worlds once we adopt a particular approach to intensionality—*The Scope Theory of Intensionality*, dating back to Montague (1970, 1973). The conclusion I will arrive at is that the scope theory's design and basic principles do not by themselves block the unattested reading, but, together with certain general principles pertaining to cumulativity we can correctly capture the empirical landscape under this theory. The discussion is not conclusive, in that it does not offer a comprehensive defense of these general principles or address the worries that pairing them with the scope theory raises. Rather, it lists some reasons why pursuing this approach, and addressing the worries to be discussed, is a promising endeavor.

I proceed as follows: Section 2 presents Schmitt's (2023) argument for the ban on pluralities of worlds in broad strokes, setting the stage for a new argument. Section 3 presents the standard picture regarding the unavailability of wide scope opaque readings, and Section 4 illustrates how different theories of intensionality attempt to block these readings. Section 5 shows how, once these theories are augmented with a cumulation mechanism they actually fail to block the unattested readings. Section 6 speculates on how the unseemly interaction between cumulation and intensionality can be blocked under the scope theory of intensionality. Section 7 concludes.

2 The ban on cumulativity with pluralities of worlds

Plural predication gives rise to certain *cumulative* inferences from truths about individuals to truths about pluralities thereof. The truth of the sentences in (3a-3c), for instance, in which the one-place predicate "*slept*" is asserted to hold of several individuals separately, entails the truth of the sentence in (3d), where the predicate is applied to a plurality of said individuals (presumably denoted by the conjunction; Link 1983). Similarly, the truth of the three sentences in (4a-4c)—where the two-place predicate "*talk to*" is asserted to hold of several pairs of individuals—suffices for us to conclude that the predicate also holds of the pair consisting of the pluralities of these individuals, as indicated by the truth of (4d) given (4a-4c).

(3) a. Sue slept.
b. Jane slept.
c. Mary slept.
d. ∴ Mary, Jane and Sue slept.

(4) a. Student 1 talked to Professor 1.
b. Student 1 talked to Professor 2.
c. Student 2 talked to Professor 3.
d. ∴ The two students talked to the three professors.

More abstractly, we can define a cumulative inference as in (5), for predicates of any arity, though we will only be concerned with one- and two-place ones.

(5) For any n-place predicate P^n, and any m n-tuples of arguments,
$<x_1^1, x_2^1, \ldots, x_n^1>, <x_1^2, x_2^2, \ldots, x_n^2>, \ldots, <x_1^m, x_2^m, \ldots, x_n^m>$
(where the lower index on every variable represents its position within its tuple, and the upper index represents the tuple of which it is a part),
the following is a cumulative inference
$<x_1^1, \ldots, x_n^1> \in P^n \wedge <x_1^2, \ldots, x_n^2> \in P^n \wedge \ldots \wedge <x_1^m, \ldots, x_n^m> \in P^n \Rightarrow$
$<x_1^1 \oplus \ldots \oplus x_1^m, x_2^1 \oplus \ldots \oplus x_2^m, \ldots, x_n^1 \oplus \ldots \oplus n_n^m> \in P^n$
(where $\oplus$, here and throughout, is the sum-formation operation, returning for any two individuals, the individual that is their mereological sum)

Schmitt's (2023) argument against the existence of world-plurality denoting expressions uses the presence of cumulative inferences to diagnose for plural denotations in a sentence. It goes roughly as follows: First, it is observed that conjunctions of propositions may partake in cumulative relations with other plurality-denoting expressions. Yet, no expressions previously argued to denote pluralities of worlds partake in cumulative relations with propositional conjunctions. Thus, if cumulativity is a reliable diagnostic for plurality (cf. Schmitt 2019), an analysis of these constructions in terms of world pluralities seems to be on the wrong track.

To see that conjunctions of propositions "count" as plural expressions for the purpose of deriving a cumulative reading, consider (6), where a plurality consisting of John and Bill is truthfully ascribed belief in a conjunction of propositions, even though neither John nor Bill can truthfully be ascribed belief in the conjunction on his own: In the given context, each individual only believes one of the conjoined propositions.

(6) a. **Context:** Sam owns a construction company. Bill and John are her clients. John believes Sam is from Texas, and has no opinion on whether she is rich, while Bill believes she is rich but does not know where she is from.

b. John and Bill believe that Sam is from Texas and that she is rich.
(Marty 2019)

Pasternak (2018) attributes the felicity of such ascriptions of "joint belief" to the fact that the beliefs of John and Bill are mutually compatible, and thus, the conjunction of their belief states entails the conjunction in the prejacent of "*believe*" in (6b). However, Marty (2019) shows that joint belief attribution subscribes to the condition on cumulativity, mentioned earlier (and made evident in (5)), according to which a relation holds cumulatively between two pluralities only if it does not "leave out" any member of either. The condition, spelled-out specifically for joint belief in (7), plays a role once the context is manipulated as follows: Suppose John believes Sam is from Texas, but Bill only believes that if Sam is from Texas, she is rich. In this context, (6b) no longer seems true, despite the fact that the conjunction of John and Bill's belief states still entails the conjunction in the complement of the attitude verb. What is different now is that the modified context violates the condition in (7), and as Marty notes, this is at least suggestive that "joint belief" reduces to a cumulative belief-relation holding between a plural subject and a conjunction of propositions.

(7) An individual plurality $x_1 \oplus \ldots \oplus x_i$ can be ascribed joint belief in a conjunction $p_1 \wedge \ldots \wedge p_j$ only if the *belief*-relation holds *cumulatively* between the plurality and the conjunction: i.e., only if each atom in $x_1 \oplus \ldots \oplus x_i$ believes at least one conjunct in $p_1 \wedge \ldots \wedge p_j$, and each conjunct in $p_1 \wedge \ldots \wedge p_j$ is believed by at least one atom in $x_1 \oplus \ldots \oplus x_i$.

Schmitt (2023) illustrates that in modal constructions previously analyzed as involving pluralities of worlds, the hypothesized world pluralities fail to relate cumulatively to conjunctions of propositions. Below, we reproduce Schmitt's illustration that *if*-clauses, analyzed as world pluralities by Schlenker (2004), Gajewski (2005), and Križ (2015), do not partake in a cumulative relation with a propositional conjunction. We then provide a parallel illustration for weak necessity modals, also recently analyzed as plurality-denoting expressions by Agha & Jeretič (2022).

On the world-plurality analysis of *if*-clauses, "*if*" is assigned, essentially, the semantics of a definite article: Given a proposition p and world argument w, it returns *the* plurality of p-worlds maximally similar to w, given a similarity ordering. The conditional as a whole then predicates its consequent proposition on the world plurality denoted by its antecedent. This is spelled-out in (8).

(8) a. $[\![if_w]\!] = \lambda p.\ \bigoplus \text{CLOSEST}(w)(p)$ (where, $\forall w, p : \bigoplus \text{CLOSEST}(w)(p)$ is the plurality of p-worlds most similar to w)

b. $[\![[if_w\ \phi]\ [\psi]]\!] = [\![\psi]\!]([\![if_w]\!]([\![\phi]\!])) = [\![\psi]\!](\bigoplus \text{CLOSEST}(w)([\![\phi]\!]))$

c. $[\![If_w$ *Abe wins, Ron will be happy*$]\!]$ = 1 *iff* Ron is happy in the (maximal) world plurality in which Abe wins and which is otherwise maximally similar to w

A conditional thus asserts that its consequent stands in a particular relation to the plurality in its antecedent; namely, the *be-true-of* or *hold-of* relation. Schmitt shows that this relation cannot hold cumulatively by examining conditionals with a conjunctive consequent. If propositional conjunctions "count" as pluralities, such conditionals are predicted to behave as if their consequent is interpreted *disjunctively*. This is because the *hold-of* relation should hold cumulatively between a conjunction and a plurality of worlds, even if each proposition in the conjunction does not hold of each world in the plurality, as long as each world in the plurality makes true *some* proposition in the conjunction (and each conjunct is true of some world). It is easy to see that such a disjunctive interpretation is unavailable for, e.g., (9b), despite the fact that the scenario in (9a) renders the utterance's cumulative paraphrase in (9c) true.

(9) a. **Context:** If Abe's company does not go bankrupt, he will hire another employee; either Sasha or Bert, but certainly not both.

b. If Abe's company does not go bankrupt, he will hire Sasha and he will hire Bert.

c. In each of the closest worlds where Abe's company does not go bankrupt, Abe hires Sasha, or he hires Bert.

The argument generalizes to any construction argued to ascribe the *hold-of* relation to a world plurality and a conjunction of propositions. Since the relation is expected to hold *cumulatively* between two pluralities, we predict the weak disjunctive interpretation to be available. If it is not, we should question the assumption that a world plurality is involved to begin with. For instance, we can apply the argument to weak necessity modals. On their analysis in Agha & Jeretič (2022), they too are definite determiners. Simplifying significantly, the modal "*should*" takes a world w and a domain D and, defined as a set of worlds that are "best" given some

contextually supplied ranking among the worlds accessible from w, and returns their maximal plurality. The modal's propositional complement is then predicated to hold of this plurality; cf. (10). And with necessity modals, too, the weak cumulative reading does not seem to arise, as illustrated by the fact that the context in (11a) seems to falsify (11b), despite verifying its cumulative paraphrase (11c).

(10) a. $[\![\text{should}_w\ D]\!] = \bigoplus D$

b. $[\![[[\text{should}_w\ D]\ \phi]]\!] = [\![\phi]\!]([\![\text{should}_w\ D]\!]) = [\![\phi]\!](\bigoplus D)$

(11) a. **Context:** The syllabus of an advanced semantics class requires that students read a paper or solve an exercise, but not both.

b. According to the syllabus, you should read a paper and solve an exercise.

c. In each "best" world given the syllabus, Abe reads a paper or solves an exercise.

From these observations (and others, which I omit here for conciseness), Schmitt (2023) concludes that overt expressions previously argued to form pluralities of worlds do not, in fact, do so.[2] However, as Schmitt notes, proponents of the world plurality hypothesis have some potential workarounds. For instance, they can stipulate that the absence of cumulative readings in (9) and (11) is due to a quirk in the syntax or lexical semantics of conditional operators and necessity modals, which prevents the grammatical devices in charge of deriving cumulativity from applying. Perhaps these elements denote world pluralities that are "inherently distributive"—either by mandatorily combining with a distributivity operator or by encoding distributivity into their lexical entries—thus requiring the world pluralities they denote to render true *every* proposition in the conjunction they combine with? In that case, the world pluralities involved are not expected to partake in cumulative relations in the first place. Alternatively, they can reject the assumption that conjunctions of propositions "count" as pluralities, and instead pursue Pasternak's (2018) analysis of examples like (6) as instances of "joint belief" rather than as instances in which a cumulative *belief*-relation holds between a plurality of individuals and a conjunction of propositions.[3]

2 The world plurality analysis in, e.g., (8) and (10) is motivated primarily by the observation that *if*-clauses and weak necessity modals give rise to *homogeneity* and *non-maximality* effects, which are attested also in the presence of expressions denoting *individual* pluralities (cf. Gajewski 2005). If Schmitt is right to reject the world plurality analyses, these shared interpretative effects require an alternative explanation. I set this issue aside for the purposes of the current paper, but note that, as Schmitt already points out (citing Burnett 2016) homogeneity and non-maximality arise even in the absence of plurality, particularly in certain degree constructions. It is possible, then, that these effects are correlated with an independent property shared by certain plural and non-plural expressions, and are thus unreliable diagnostics for plurality, in contrast to cumulativity.

3 Of course, in that case, one would have to explain why such cases are sensitive to the condition on

Schmitt lists some conceptual arguments against this approach, noting that it would make plurality in the world domain undiagnosable and therefore, the world plurality hypothesis unfalsifiable. Furthermore, if a world plurality analysis is appealed to in the first place to model certain interpretative parallels to individual plurality, introducing inherent asymmetry raises non-trivial questions; in particular, why should only *some* properties that characterize individual plurality carry over to the world domain if not all of them do?

But even setting aside these conceptual reservations, the argument to be presented next illustrates that stipulations of this sort are also *empirically* inadequate. I will show that the scope of Schmitt's generalization—that worlds do not partake in cumulative readings—goes beyond the particular expressions she examines, and thus that deriving the absence of a cumulative reading in (9) or (11) by analyzing *if*-clauses and necessity modals in a particular way, or by rejecting Schmitt's assumption about propositional conjunctions, simply would not do. The argument takes as its starting point the observation that a certain scopal configuration seems systematically unavailable for DPs in intensional context, an observation that we turn to next.

3 On the non-existence of wide scope opaque readings

DPs in intensional environments may vary both in the scope they take relative to an intensional operator, and in whether their NP restrictors are evaluated *opaquely*, in the worlds quantified by that operator, or *transparently*, in the matrix world of evaluation. However, these two dimensions of variation are not independent: Wide quantificational scope for a DP blocks opaque evaluation of its restrictor.

This is often illustrated with the well-known sentence in (12), due to Fodor (1970). The sentence contains an existential DP "*an inexpensive coat*" and the attitude predicate "*want*", which we take to be a universal quantifier over possible worlds that satisfy Mary's desires (Hintikka 1962).

(12) Mary wants to buy an inexpensive coat. (Fodor 1970)

(13) a. $\lambda w.\ \forall w' \in \text{DESIRE}_{Mary,w}\colon \exists x\colon \mathit{cheap\text{-}coat}_{w'}(x) \wedge \mathit{buy}_{w'}(\mathit{Mary}, x)$ *(de dicto)*

b. $\lambda w.\ \exists x\colon \mathit{cheap\text{-}coat}_{w}(x) \wedge \forall w' \in \text{DESIRE}_{Mary,w}\colon \mathit{buy}_{w'}(\mathit{Mary}, x)$ *(de re)*

c. $\lambda w.\ \forall w' \in \text{DESIRE}_{Mary,w}\colon \exists x\colon \mathit{cheap\text{-}coat}_{w}(x) \wedge \mathit{buy}_{w'}(\mathit{Mary}, x)$ *(third)*

d. $\lambda w.\ \exists x\colon \forall w' \in \text{DESIRE}_{Mary,w}\colon \mathit{cheap\text{-}coat}_{w'}(x) \wedge \mathit{buy}_{w'}(\mathit{Mary}, x)$ *(R4)*

cumulative relations, which requires such relations to hold between two pluralities without leaving out any members of those pluralities.

	Transparent	Opaque
Wide-scope (*specific*)	✓	✗
Narrow-scope (*non-specific*)	✓	✓

Table 1 DPs in intensional contexts

The sentence has three readings: (i) A *de dicto* reading, true as long as the purchase of some inexpensive coat or other meets Mary's desires. (ii) A *de re* reading, verified by there being a *specific* inexpensive coat Mary wants to buy, possibly without knowing that it is inexpensive, or even a coat. And (iii) a *third* reading, verified by the existence of several objects such that buying either would satisfy Mary, which all happen to be inexpensive coats (again, possibly unbeknownst to Mary). The *de dicto* reading is often analyzed as involving narrow scope for the existential DP relative to the attitude with opaque evaluation of its restrictor (13a), the *de re* reading pairs wide scope with transparent evaluation (13b), and the third reading involves narrow scope with transparent evaluation (13c).

A reading that (12) does *not* seem to have, on the other hand, is the one formalized in (13d), which would be derived had the existential DP outscoped the attitude verb while its restrictor were evaluated opaquely, and has been called in the literature its *wide scope opaque reading* or its *fourth reading*. Going forward, I will adopt the latter term, henceforth **R4**, which is more theory neutral than the former. R4 is verified by the existence of a particular object that Mary wants to buy, thinking that it is inexpensive, even if that object is actually quite expensive. To illustrate the non-existence of R4, Keshet (2008) notes that the R4 of (12) is equivalent to (14a). But (14a) is not a possible interpretation of (12), as illustrated by the fact that when pronominal reference is used to force wide scope for the DP over the attitude verb, as in (14b), opaque evaluation of the term "*inexpensive*" is rendered unavailable.

(14) a. There's a coat that Mary wants to buy. She thinks it is inexpensive. But really, it is quite expensive.

b. Mary wants to buy an inexpensive coat. #But really, it is quite expensive.

Table 1 thus represents the picture accepted by much of the semantic literature, according to which wide scope and opaque evaluation are incompatible with each other.[4] Indeed, contemporary discussions of intensionality and scope often treat the ability to capture the empirical landscape exactly as reflected in the table—i.e.,

4 My presentation in this section is somewhat anachronistic: In fact, Fodor (1970) originally claimed that R4 is attested, and offered (12) as an illustration of a sentence that does have this reading. Subsequent research rejected Fodor's original judgment (Ioup 1975; Keshet 2008, 2010; von Fintel & Heim 2011; Keshet & Schwarz 2019), with Keshet 2008 offering the data above as an argument in favor of this rejection. Szabó (2010, 2011) argues that the rejection of Fodor's main claim, that

blocking the derivation of R4, the reading that would be derived by combining wide scope and opaque evaluation—as a benchmark for an empirically adequate theory of intensionality (e.g., Demirok 2019; Elliott 2023; Mirrazi 2024). Next, I illustrate how two families of theories of intensionality are designed to block R4, focusing on *The Scope Theory of Intensionality* and its modern rendition in Keshet's (2008; 2010) termed *Split Intensionality*, and on *The World Variable Framework* whose modern incarnation was developed by Percus (2000), building on a long tradition in modal logic (cf. among many others Tichý 1971; Gallin 1975; van Benthem 1977; Groenendijk & Stokhof 1982; Zimmermann 1989; Cresswell 1990/2012).

There are two reasons to focus on these theories in particular: First, they are very popular, with the world variable framework in particular constituting the go-to formal framework in which intensional constructions are analyzed (Yanovich 2015: 66), and with Keshet's split intensionality system treated as "the state of the art scope theory" (Elliott 2023: 334). More importantly, for the purposes of this paper, the two approaches represent here two *families* of approaches, differing in their assumptions about the range of elements that object-language expressions may denote, and my eventual goal is to show that regardless of such assumptions, both families suffer from an overlooked overgeneration problem.[5]

4 Blocking wide scope opaque readings

The two systems to be considered here were both developed in response to the shortcomings of the classical *Scope Theory of Intensionality* (henceforth, **STI**; Montague 1970, 1973; Ladusaw 1977). STI maintains that the intensional status of a world-dependent element is determined by the nearest intensional operator ω above it at LF; i.e., that every world-dependent element in the c-command domain of ω is interpreted opaquely relative to ω. Paired with a syntactic approach to scope, STI predicts a biconditional dependence between quantificational scope and intensional status: as shown in Figure 1, any element under ω will receive narrow-scope and opaque evaluation relative to ω, and any element taking scope above ω will receive transparent evaluation. As desired, biconditional dependence (seemingly) rules out

R4 exists, was too hasty, presenting both new data purportedly illustrating the existence of R4, and an argument that this data is indicative of logical forms in which an intensional operator scopes between the determiner and restrictor of a single DP. However, Santorio (2013); Benbaji (2021); Benbaji-Elhadad (2026b,c); Mayr & Schmitt (2026) show that Szabó's theoretical conclusion is too broad, overgenerating unattested instances of R4, and posit alternative more restrictive mechanisms to derive the instances that Szabó shows are attested. Since this paper is concerned with showing that, even without positing a mechanism that lets a determiner split from its restrictor, we already overgenerate unattested cases of R4 via cumulation, I set this other debate aside.

5 The presentation of these approaches below closely follows Keshet & Schwarz 2019, and the figures are reproduced from Benbaji-Elhadad 2026b.

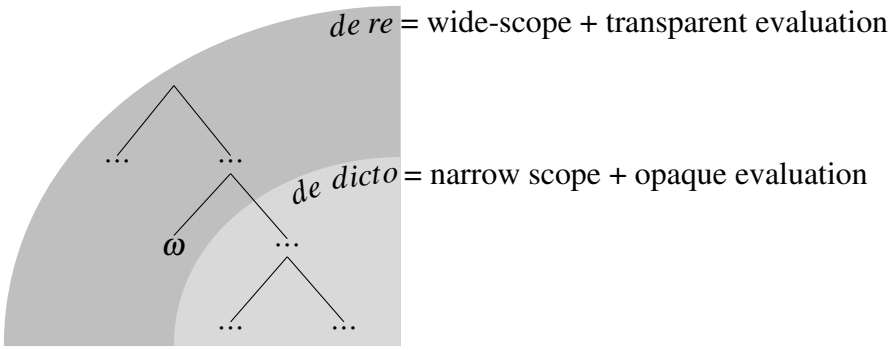
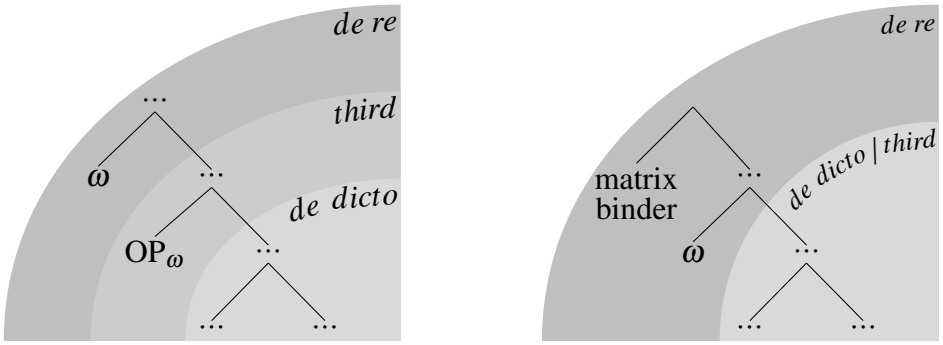


Figure 1 STI: wide-scope entails transparent evaluation; narrow-scope entails opaque evaluation



(a) Split intensionality (b) World variables

Figure 2 Modifications of STI: wide-scope entails transparent evaluation while narrow-scope is compatible with transparent or opaque evaluation

R4, which requires wide scope and opaque evaluation. However, it also rules out the narrow-scope transparent *third* readings. STI, then, fails to capture the empirical landscape spelled-out in Table 1; the feature of the theory in charge of blocking unattested cases of R4—namely, the biconditional dependence of scope and intensional status—also leads to the undergeneration of attested ones.

Keshet (2008, 2010) modifies STI by partially divorcing scope from intensional status. On his modified version, being below ω at LF entails narrow-scope, but is not sufficient for opacity. DPs are opaque only if they reside below *some other operator*, call it OP_ω, itself situated below ω at LF. In the resulting system of *split intensionality*, there is a region under ω where elements receive transparent evaluation; namely, the region between ω and OP_ω. As illustrated in Figure 2-a, as in the case of STI, syntactic position still fully determines a DP's intensional status in the

modified system, but STI's biconditional dependence becomes a conditional one: wide-scope entails transparency, but not vice versa.

The same conditional dependence is the result of Percus's (2000) more radical departure from STI. In his system, world-dependent lexical items are functions whose input is a world-argument, saturated by silent object-language *world-variables* at LF. The restrictor of every DP is assumed to contain such a variable as its argument, which can be bound either by an intensional operator above it or by a binder posited at the matrix level, with the identity of the binder determining the intensional status of the DP. When a world-variable is bound by the matrix binder, the restrictor is transparent, and it is opaque when its variable is bound by the intensional operator above it. As shown in Figure 2-b, here, too, residing under ω does not determine a DP's intensional status, but being outside it does: ω cannot bind the world-pronoun of a DP above it, which must thus be evaluated transparently.

Both approaches generate the *third* reading by positing a mechanism that allows elements in the c-command domain of an intensional operator ω to escape opacity relative to ω. However, there is no mechanism in place that would allow an element outside the c-command of ω to nevertheless be opaque. Thus, R4 is still ruled out by both modifications of STI. Or is it?

5 Failing to block wide scope opaque readings

Next, I illustrate that certain cases of R4 can nevertheless be derived within the standard theories of intensionality schematized above, despite those features of the theories intended to block them. As noted, these features ensure that a DP will not be able to occupy a position above an operator that in turn determines the intensional status of its restrictor. However, what I show next is that given an independently needed cumulation mechanism, one can derive for certain sentences, under either theory, a reading that entails R4, even when those sentences are mapped to LFs in which the relevant DP is positioned *below* an intensional operator. We thus predict such sentences to be true in scenarios that are verified by R4, contrary to fact.

The sentences I have in mind contain two existential quantifiers: A plural indefinite DP, quantifying existentially over individual pluralities, and a possibility modal, quantifying existentially over possible worlds. One example, recall, is (15), which contains the epistemic modal "*might*" and the plural indefinite "*two victims*".

(15) Two victims might be trapped under the rubble.

The main empirical observation I will be concerned with consists of the fact that (15) does not seem to be verified by the existence of two individuals who could *each* be a victim trapped under the rubble, even when there are no two individuals who could *both* be victims trapped under the rubble. This is the state of affairs in

the scenario in (16) which, indeed, does not seem to verify (15).

(16) **Collapse scenario:** A building collapsed, with only the security guard on the night shift present inside. There are only two people who are ever on duty at night, but we are not sure which of them picked up the shift on the night of the collapse. Thus, for all we know, each of them *might* be a victim of the collapse that is trapped under the rubble, but we are certain that it is not the case that both are.

R4 of (15) is predicted to be *true* in this scenario, as it pairs wide quantificational scope for "*two victims*" relative to the modal with opaque evaluation of the restrictor, and is thus paraphrasable as follows: There are two individuals who could each be a victim trapped under the rubble.[6] Therefore, we must ensure that R4 is not derivable for (15).[7]

In the remainder of this section, I will show how each of the approaches to intensionality presented above, augmented with a cumulation mechanism, derive a reading of (15) that is true in the scenario above which, as a matter of fact, falsifies it. To that end, I will adopt a working hypothesis regarding how cumulative inferences come about. I will then show how with this hypothesis in place, we wrongly predict the R4 of a sentence like (15) to be attested.

6 I introduce the distributivity operator "*each*" in the paraphrase, which indicates that the predicate in the scope of the plural indefinite quantifier must hold of every atomic individual in the pluralities being existentially quantified over. This is because any two existential quantifiers are scopally-commutative without distributivity; i.e., the expression obtained from scoping one above the other entails the expression obtained from reversing this scopal configuration, as long as the scope of the inner quantifier remains constant (for any scope ϕ, $[\exists x \exists y \phi(x,y)] \Leftrightarrow [\exists y \exists x \phi(x,y)]$). Therefore, we might expect readings that differ only in the scope they assign to one existential over another to be indistinguishable. In particular, the R4 of (15) without distributivity, according to which for two individuals there is a world in which they are (both) trapped under the rubble, seems to be equivalent to, and thus indistinguishable from, its *de dicto* reading, according to which there is a world in which two victims are trapped under the rubble. Whether or not commutativity should be assumed to hold when one of the existential quantifiers involved is a modal, as is the case here, is subject to some debate in the study of modal logic (see, for instance Williamson's (2013) discussion of the *Barcan formula* and its converse, due to Barcan (1946)), and the introduction of distributivity here allows us to side-step this debate altogether.

7 We can reproduce Keshet's (2008) argument, illustrating that Fodor's (1970) sentence in (12) lacks a R4, for (15) as well. If the sentence had a reading equivalent to the R4 paraphrase, a pronoun in a succeeding sentence should be able to refer back to those individuals individuated by the wide-scope existential DP, who are potential but not actual victims. The contrast below, modeled after (14) illustrates that this is not the case.

(i) a. There are two individuals who could each be a victim trapped under the rubble. However, they are not both victims.

b. Two victims might be trapped under the rubble. #However, they are not both victims.

However, since that discussion will naturally involve quite a bit of technicalities, let us first briefly formulate the essential points to be made below in non-technical terms. First, in both the world variable system and the system of split intensionality, the predicate "*victims*" can contribute to the semantic derivation a relation, true of an individual and a world if that individual is a victim in that world. I will apply a cumulation mechanism to that relation, rendering it true of two *pluralities*, of individuals and worlds, just in case: (i) for each part of the individual plurality there is a world in the world plurality, in which it is a victim, and (ii) for each world in the world plurality there is a part of the individual plurality that is a victim in it. Then, I will also use cumulation to allow the modal "*might*" to quantify existentially over pluralities of worlds. Together, this will render (15) true by the existence of one world in which some individual is a victim, and another world in which some other individual is a victim, which is, in turn, precisely what goes on the collapse scenario. Thus, (15) is predicted by both systems to be true in a scenario that exclusively verifies its R4, contrary to fact.

5.1 Assumptions about the cumulation mechanism

Recall that cumulative inferences are the inferences of the (d)-sentences below, drawn from the (a-c)-sentences. Given that we seem justified in drawing such inferences, we need a mechanism in place that will render the (d)-sentences true, in case the assumptions in (a-c) are.

(17) a. Sue slept.
b. Jane slept.
c. Mary slept.
d. ∴ Mary, Jane and Sue slept.

(18) a. Student 1 talked to Professor 1.
b. Student 1 talked to Professor 2.
c. Student 2 talked to Professor 3.
d. ∴ The two students talked to the three professors.

Let me mention two such mechanisms in particular. On one approach, the extensions of predicates are closed under sum-formation as part of their *lexical meaning*; e.g., the denotation of a verb like "*slept*" consists of the set of all individuals who slept, as well as of any individual formed by summing up any *n* sleepers (cf. Scha 1981; Krifka 1992; Landman 1996; Winter 2000; Kratzer 2003, 2008, *ao*). In other words, it is part of the lexically specified meaning of predicates that they give rise

to cumulative inferences. This explains, for instance, why (17d) is true when (17a-17c) are; since the atomic individuals Sue, Jane and Mary are in the extension of "*slept*", then *ipso facto* so is the individual formed by summing them up.

On another approach, closure of predicate extensions under sum-formation is not a lexical property, but the result of applying cumulation operators to predicates at LF (cf. Link 1983; Krifka 1986; Sternefeld 1998; Beck & Sauerland 2000, *ao*). These include Link's (1983) "star-operator" $*$, which applies to one-place predicates, Krifka's (1986) extension of it to two-place ones, "double-star" $**$, and any further extensions to predicates of higher-arity (which we henceforth ignore for simplicity). Type-flexible versions of these operators are defined in (19a-19b), with the effect of the latter on the two-place predicate "*talk to*" exemplified in (20), illustrating why the cumulative inference in (18) goes through.

(19) a. $[\![*]\!] = \lambda \mathcal{F}_{\langle\sigma,t\rangle} \lambda x_{\sigma}.\ \exists \mathcal{F}'_{\langle\sigma,t\rangle} : [\forall x' : \mathcal{F}'(x') \rightarrow \mathcal{F}(x')] \wedge x = \bigoplus\{x \mid \mathcal{F}'(x)\}$

b. $[\![**]\!] = \lambda \mathcal{R}_{\langle\sigma,\langle\tau,t\rangle\rangle} \lambda x_{\sigma} \lambda y_{\tau}.\ \exists \mathcal{R}'_{\langle\sigma,\langle\tau,t\rangle\rangle} : [\forall x', y' : \mathcal{R}'(x', y') \rightarrow \mathcal{R}(x', y')] \wedge$
$x = \bigoplus\{x \mid \exists y : \mathcal{R}'(x, y)\} \wedge y = \bigoplus\{y \mid \exists x : \mathcal{R}'(x, y)\}$

(20) a. $[\![\text{talk to}]\!] = \{\text{<Student 1,Prof 1>}, \text{<Student 1,Prof 2>}, \text{<Student 2,Prof 3>}\}$

b. $[\![**]\!]([\![\text{talk to}]\!]) = \{\text{<Student 1,Prof 1>}, \text{<Student 1,Prof 2>},$
$\text{<Student 2,Prof 3>}, \text{<Student 1,Prof 1}\oplus\text{Prof 2>},$
$\text{<Student 1}\oplus\text{Student 2,Prof 1}\oplus\text{Prof 3>},$
$\text{<Student 1}\oplus\text{Student 2,Prof 2}\oplus\text{Prof 3>},$
$\text{<Student 1}\oplus\text{Student 2,Prof 1}\oplus\text{Prof 2}\oplus\text{Prof 3>}\}$

The operator approach *subsumes* the lexical one in that any cumulative inference derived by assuming predicates are "born" with their arguments closed under sum-formation, could also be derived by assuming closure under sum-formation occurs in the syntactic derivation itself. In fact, the main claim to fame of the operator approach is that it allows cumulation of *derived* predicates—i.e., of predicates denoted by non-terminal nodes in a syntactic structure—and not just of lexical predicates in the terminal "leaf" nodes of the syntactic tree (cf. Beck & Sauerland 2000). The operator approach is thus more expressive than the lexical one, and so are other alternatives to lexical cumulativity on the market, which were all proposed as ways to derive cumulative inferences that are undergenerated by the lexical approach.[8] These various proposals all agree that we need a mechanism that is *at least* as expressive as the operator approach to account for the range of attested cumulative inferences, with some—notably Schmitt (2019)—faulting the operator approach for not being expressive *enough*.

8 These include, for instance, Schmitt's (2019; 2020) theory of *plural projection*, and the account of cumulation in the framework of *event semantics* (Schein 1993; Landman 2000; Kratzer 2003, 2008; Ferreira 2005).

I will not discuss alternative approaches any further for a simple reason: The goal of this paper is to illustrate that augmenting our intensional semantics with a cumulation mechanism leads to the overgeneration of an unattested reading; namely, R4. If I can show that overgeneration arises even by adopting the cumulation mechanism whose expressive power is generally agreed in the literature to set a lower bound on what is required to correctly capture the empirical landscape—namely, the operator approach—then this should suffice for the reader to extrapolate that any more expressive mechanism would give rise to the same problem.

5.2 Deriving R4 as a cumulative inference

Now let us be slightly more explicit about the implementation of the intensional systems reviewed in Section 4. Recall that we encountered two kinds of frameworks. On the one hand are scope-based theories, like the classical scope theory of intensionality (STI) and its modification in the form of Keshet's (2008) split intensionality, in which both the scope of a DP relative to an intensional operator and the intensional status of its restrictor are determined by the DP's position at LF. On the other hand are world variable systems, in which scope is fully determined structurally, but intensional status is not.

The main dividing line between the two families of approaches has to do with how they determine the evaluation world of a world-dependent lexical item. In variable systems, this is done by binding object language variables that are fed as arguments to such lexical items. In scope-based systems, the evaluation world of a lexical item is determined by its position relative to certain operators in the structure. From a technical standpoint, the two approaches differ in what they *minimally* need their type inventory to include; or, more specifically, in whether or not they need a primitive type s of possible worlds as part of their type inventory.[9] Scope-based systems can do without worlds in their type inventory; they need not posit (21d) as a clause in their recursive definition of semantic types, appealing to possible worlds only when constructing intensions (21e). Variable-based systems, on the other hand, must include both (21d) and (21e), as they use syntactic variables ranging over elements of type s as evaluation indices.[10]

(21) A recursive definition of semantic types:

a. e, for individuals, is a type.

b. t, for truth-values (1 or 0), is a type.

9 See Zimmermann 2018 for a useful discussion.

10 Notably, it is whether a system can do without positing elements of type s that sets scope and variable systems apart. This is because, while scope systems *need not* accommodate elements of type s, they *can* do so. I thank REMOVED (p.c.) for a discussion of this point.

c. If σ and τ are types, then so is $\langle \sigma, \tau \rangle$.

d. s, for possible worlds, is a type.

e. If σ is a type, then so is $\langle s, \sigma \rangle$; i.e., the type of intensions (functions from possible worlds of type s to elements of type σ).

f. Nothing else is a type.

Interestingly, despite their differences, there is a particular feature that unites the world variable framework and split intensionality, setting them both apart from STI: While these modifications of STI both allow open-class lexical items or particular nodes at LF that dominate them to denote intensions, classical STI does not. Next, I will show that being able to close certain intensions under sum-formation is the core ingredient in the derivation of unattested R4s, and then explore whether adopting STI, which does not make available intension-denoting nodes at LF, can block the unattested reading with less stipulations than the other systems require. For expository purposes, I will first show how one can derive R4 in a variable system and in the system of split intensionality, and only then proceed to examine how STI fares in comparison. This is, of course, not the "textbook" way; usually STI and its shortcomings are presented, and modifications are then introduced as offering a way to circumvent these shortcomings (e.g., von Fintel & Heim 2011). Since I will eventually explore a tentative defense of STI, it makes sense to do things in the opposite way here; namely, to discuss that system after demonstrating the shortcomings of its alternatives.

5.2.1 World variables

In a variable system, world-dependent lexical items are functions whose input is a world-argument. So nouns and one-place predicates like "*victim*" or "*trapped*" denote a kind of intension; namely a relation between worlds and individuals (22a-22b). The existential modal "*might*" combines with its own world-argument, and with a contextually supplied accessibility function that maps that world-argument to a set of worlds accessible from it (Kratzer 1977, 1981), returning a generalized quantifier over worlds (22c); a function true of any proposition if there is an accessible world of which that proposition holds. On the other hand, quantificational determiners, assumed here to include numerals like "*two*" (22d),[11] are world-independent; i.e., they do not take worlds as arguments, nor are their arguments themselves intensional.

(22) a. $[\![\text{victim}]\!] = \lambda w \lambda x.\ x$ is a victim in w (shorthand, victim(x, w))

11 Nothing in what follows rests on this assumption; the argument to be made below is fully compatible with an analysis of numerals as predicative elements that appear under a silent existential quantifier.

b. $[\![\text{trapped}]\!] = \lambda w \lambda x.\ x$ is trapped in w (shorthand, trapped(x, w))

c. $[\![\text{might}]\!] = \lambda w \lambda f_{\langle s,st\rangle} \lambda p_{\langle s,t\rangle}.\ \exists w' \in f(w) : p(w')$

d. $[\![\text{two}]\!] = \lambda P_{\langle e,t\rangle} \lambda Q_{\langle e,t\rangle}.\ \exists x : |x| = 2 \wedge P(x) \wedge Q(x)$
(where $|x|$ refers to the number of atomic elements in x)

In this system, following Heim & Kratzer 1998, sister nodes in a syntactic structure combine via the function application rule in (23a). FA allows a function-denoting element to combine with its syntactic sister, as long as its sister denotes an element in the domain of the relevant function. However, given that lexical elements like the noun in (22a) denote intensions, whereas determiners like (22d) that take them as complements have only extensional arguments in their domains, something more must be said about how determiners combine with their complements, and more generally, how world-dependent items combine with items that denote functions with extensional domains. Potential type-mismatch is avoided in this case by assuming that world-arguments are always saturated by variables in the object-language, which are effectively, pronouns ranging over possible worlds.

These pronouns do not remain free. Rather, each pronoun is bound by one of multiple λ-binders, positioned immediately below any intensional operator and in a position outscoping the entire matrix sentence, with the aid of two additional composition rules in (23b) and (23c).[12]

(23) a. **Function Application (FA):** If α is a node with daughters $\{\beta, \gamma\}$, then if $[\![\beta]\!]$ is a function whose domain contains $[\![\gamma]\!]$, $[\![\alpha]\!] = [\![\beta]\!]([\![\gamma]\!])$.

b. **Pronoun Rule (PR):** If α is a pronoun or variable carrying an index i, then for any assignment function g from indices to denotations, $[\![\alpha]\!]^g = g(i)$

c. **Predicate Abstraction (PA):** If α is a node with daughters $\{\beta, \gamma\}$, and β dominates only an indexed operator λ_i, then $[\![\alpha]\!] = \lambda x.\ [\![\gamma]\!]^{[i \to x]}$

Figure 3 provides the structure that would derive, in this system, the *de dicto* reading of (15), according to which it might be the case that two victims are trapped under the rubble. The world pronouns adjoining to the predicates "*victims*" and "*trapped*" in that structure are co-indexed with, and therefore bound by, the λ-operator under the existential modal "*might*". The node labeled TP in the figure thus denotes the proposition in (24), true of a world if there are two trapped victims in that world, and once the modal combines with (24) the entire sentence asserts that there is (at least) one accessible world in which there are two trapped victims.

12 Given that we will be assuming there are pronouns and variables of different types (for instance, variables that range over accessibility relations that are arguments of modals), these rules should apply generally to variables of any type. We abstract over this in the definitions below by describing an assignment g as a function from indices to denotations in general, but see Heim & Kratzer 1998: pg. 213 for a more rigorous definition.

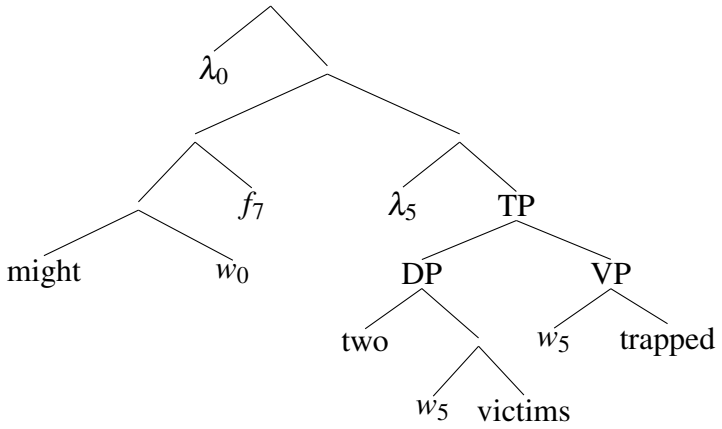


Figure 3 LF that derives the *de dicto* reading of (15), with world variables

(24) $\lambda w.\ \exists x : |x| = 2 \ \wedge\ \text{victims}(w,x) \ \wedge\ \text{trapped}(w,x)$

To derive the unattested R4 of (15)—or rather, a reading of (15) that is strictly weaker than that R4, and thus wrongly predicted to be true in a scenarios like (16)—one only has to insert cumulation operators in the right places in the structure of the *de dicto* reading above. This is done in Figure 4 below, where the two-place relations "*victims*" and "*trapped*" as well as the contextually supplied accessibility relation that is an argument of the modal, combine with the pluralizing "double-star" $**$ (19b) before merging with anything else. After combining with $**$, the predicate "*victims*" denotes a relation true of a plurality of worlds w and a plurality of individuals x, just in case for every individual part of x there is a world part of w in which that part is a victim, and for every world part of w there is an individual part of x that is a victim in that world (25a). The same holds, *mutatis mutandis*, for the predicate "*trapped*" (25b).

The contextually supplied accessibility relation, designated as f_7 in Figure 4, denotes some function of type $\langle s, st\rangle$ mapping any world to the set of worlds accessible from it. Applying a sum-formation operator to that function will result in a mapping as schematized in (25c); i.e., in a function mapping any world or plurality of worlds w to a set containing all worlds accessible from w and their sums (if w is a single world), or all worlds accessible from parts of w and their sums (if w is a plurality). Note that in the denotation of "*might*" in (22c), the accessibility function is provided with the modal's evaluation world as its first argument, with the resulting set of worlds restricting the existential quantifier introduced by the modal. If we assume that any modal's world argument is always saturated by a single world, *not* a plurality, we can simplify things by treating the pluralized accessibility function as one that maps any world to the set of worlds accessible from it once that set is

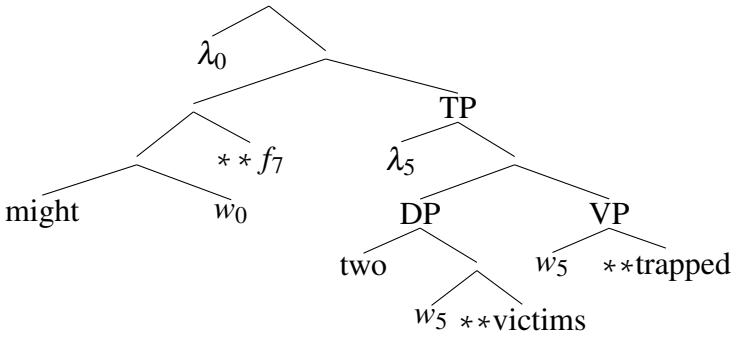


Figure 4 LF that derives R4 for (15), with world variable

itself closed under sum-formation.

(25) a. $[\![**\text{victim}]\!] = \lambda w \lambda x.\ [\forall y \leqslant x : \exists w' \leqslant w : \text{victim}(w', y)] \wedge [\forall w' \leqslant w : \exists y \leqslant x : \text{victim}(w', y)]$

b. $[\![**\text{trapped}]\!] = \lambda w \lambda x.\ [\forall y \leqslant x : \exists w' \leqslant w : \text{trapped}(w', y)] \wedge [\forall w' \leqslant w : \exists y \leqslant x : \text{trapped}(w', y)]$

c. For any function f of type $\langle s, st \rangle$, $[\![**]\!](f)$ is the following function:
$\lambda w \lambda w'.\ [\forall w'' \leqslant w : \exists w''' \leqslant w' : f(w'', w''')] \wedge [\forall w''' \leqslant w' : \exists w'' \leqslant w : f(w'', w''')]$

The derivation then proceeds as in (26). The TP in Figure 4 denotes the proposition in (26c), true of any world or plurality thereof w if there are two individuals each of whom is a trapped victim in some part of w, as long as in every part of w one of these individuals is a trapped victim.

(26) a. $[\![\text{DP}]\!]^g = \lambda f.\ \exists x : f(x) \wedge |x| = 2 \wedge [\forall y \leqslant x : \exists w' \leqslant w_5 : \text{victim}(w', y)] \wedge [\forall w' \leqslant w_5 : \exists y \leqslant x : \text{victim}(w', y)]$

b. $[\![\text{VP}]\!]^g = \lambda x.\ [\forall y \leqslant x : \exists w' \leqslant w_5 : \text{trapped}(w', y)] \wedge [\forall w' \leqslant w_5 : \exists y \leqslant x : \text{trapped}(w', y)]$

c. $[\![\text{TP}]\!]^g = \lambda w.\ \exists x : |x| = 2 \wedge [\forall y \leqslant x : \exists w' \leqslant w : \text{victim}(w', y)] \wedge [\forall w' \leqslant w : \exists y \leqslant x : \text{victim}(w', y)] \wedge [\forall y \leqslant x : \exists w' \leqslant w : \text{trapped}(w', y)] \wedge [\forall w' \leqslant w : \exists y \leqslant x : \text{trapped}(w', y)]$

d. $[\![\text{might}_{w_0}\ **f_7]\!]^g = \lambda q.\ \exists w \in [\![**]\!]g(7)(w_0) :\ q(w)$

e. $[\![\text{Figure 4}]\!]^g = \lambda w.\ \exists w' \in [\![**]\!]g(7)(w) : \exists x : |x| = 2 \wedge [\forall y \leqslant x : \exists w'' \leqslant w' : \text{victim}(w'', y)] \wedge [\forall w'' \leqslant w' : \exists y \leqslant x : \text{victim}(w'', y)] \wedge [\forall y \leqslant x : \exists w'' \leqslant w' : \text{trapped}(w'', y)] \wedge [\forall w'' \leqslant w' : \exists y \leqslant x : \text{trapped}(w'', y)]$

Having combined with its world argument and accessibility function (26d), the modal then takes this proposition as its input, asserting it holds of some accessible world or plurality thereof (26e).

Now, consider again the collapse scenario in (16), relative to which (15) is uttered, and assume (15) does, in fact, have the truth conditions we derive for it in (26e). In that scenario, each of two individuals could be victims trapped under the rubble. This means that there is one accessible world in which one individual is a trapped victim, and another accessible world in which another individual is a trapped victim. Let us call these accessible worlds w and w', and those individuals x and x'. In this scenario, then, the following holds:

(27) a. $< w,x >, < w',x' > \in$ $[\![\text{victims}]\!]$
b. $< w,x >, < w',x' > \in$ $[\![\text{trapped}]\!]$
c. $< w \oplus w', x \oplus x' > \in$ $[\![**\text{victims}]\!]$
d. $< w \oplus w', x \oplus x' > \in$ $[\![**\text{trapped}]\!]$

Since it is stipulated that both w and w' are accessible in the world in which the scenario unfolds, the plurality $w \oplus w'$ is in the set that restricts existential quantification introduced by the modal, given that we use $**$ to close this set under sum-formation. Thus, the truth conditions we derive for (15) in (26e) are verified in the collapse scenario: The plurality of accessible worlds $w \oplus w'$ witnesses the existential quantification over worlds in (26e), and the two-sized plurality of individuals $x \oplus x'$ witnesses the existential quantifier over individual pluralities.

The reading we derive is strictly weaker than what we define as R4 of (15); i.e., that reading that would be derived from the unavailable scopal configuration, in which the determiner "*two*" outscopes the modal, while its restrictor "*victim*" is opaque relative to the modal. On this reading, the sentence is true if two individuals are such that for each there is a possible world in which they are a victim trapped under the rubble. According to the truth conditions in (26e), the sentence is true if there are two individuals and two worlds, such that each individual is a victim in at least one of these worlds, and is trapped in at least one of these worlds, and each of these worlds is such that at least one of the individuals is a victim in it, and at least one individual is trapped in it. This leaves open the possibility that each of the two individuals existentially quantified over is a victim in one world and trapped in another. However, given that the collapse scenario in (16) verifies the strong R4 of (15), it also verifies the weaker reading we derive via cumulation.

Within the world variable framework, then, one can derive a reading for (15) that is predicted to be true in a scenario like (16). Furthermore, this can be done by simply inserting cumulation operators in the structure that derives the *de dicto* reading for the sentence, without actually assigning wide scope for the DP "*two victims*". Thus, the feature of the world variable framework in charge of blocking

the derivation of R4 in the first place—namely, the feature preventing elements not in the scope of an intensional operator from receiving opaque evaluation relative to that operator—does not actually do what it is intended to do. Or, rather, while it blocks a particular structure that would give rise to a reading that is true in an exclusively-R4-verifying scenario, it does not block all possible structures that give rise to such a reading.[13]

5.2.2 Split intensionality

I now turn to scope theories of intensionality. As mentioned earlier, I will focus on Keshet's (2008; 2010) system of split intensionality, but it will be useful to present in more detail how that system was developed as a modification of the classical scope system.

The paradigmatic scopal approach is Montague's (1970; 1973) *Scope Theory of Intensionality* (STI), a modernized textbook version of which is provided in Heim & Kratzer 1998; von Fintel & Heim 2011. There, no linguistic expressions denote entities of type s; i.e., there are no world pronouns. Rather, a world parameter is tacked onto the model's interpretation function in addition to the assignment parameter g—i.e., $[\![\cdot]\!]^{w,g}$—which maps any item to its *extension* in w. Thus, an NP like "*victims*" is mapped to the set of victims in the world of evaluation (28a). Declarative sentences are mapped to truth values, 1 if true in the evaluation world and 0 otherwise (28b). Finally, intensional operators like modals and attitudes are taken to quantify over interpretative parameters, shifting the world parameter of elements in their scope (28c). The modal in (28c) takes a contextually supplied accessibility function as before, returning a function from propositions to truth values.

(28) a. $[\![\text{victims}]\!]^{w,g} = \lambda x.\ \text{victim}(w,x)$

b. $[\![\text{Two victims are trapped}]\!]^{w,g} = 1$ *iff* $\exists x : |x| = 2 \wedge \text{victim}(w,x) \wedge \text{trapped}(w,x)$

c. $[\![\text{might } f_7]\!]^{w,g} = \lambda p_{\langle s,t\rangle}.\ \exists w' \in g(7)(w) : p(w') = 1$

Note that since the interpretation function is now parameterized, we have to reformulate our composition rules as in (29-31).

(29) **Function Application (FA):** If α is a branching node with daughters $\{\beta, \gamma\}$, then for any world w, if $[\![\beta]\!]^w$ is a function whose domain contains $[\![\gamma]\!]^w$, then $[\![\alpha]\!]^w = [\![\beta]\!]^w([\![\gamma]\!]^w)$.

13 This suggests that using the term *wide scope opaque* to refer to the relevant unattested reading is misleading: The reading achieves the same effect as one would achieve by assigning wide scope to the numeral "*two*" while evaluating the NP "*victims*" opaquely, but it is shown above that the reading can also be derived without this non-standard scopal configuration.

(30) **Pronoun Rule (PR):** If α is a pronoun or variable carrying an index i, then for any world w and assignment function g from indices to denotations, $[\![\alpha]\!]^{w,g} = g(i)$

(31) **Predicate Abstraction (PA):** If α is a branching node with daughters $\{\beta, \gamma\}$, and β dominates only an indexed lambda-operator λ_i, then for any world w, $[\![\alpha]\!]^{w} = \lambda x.\ [\![\gamma]\!]^{w,[i \to x]}$

Now, consider again (15), and suppose we assign to that sentence the LF in (32). In this structure, the modal takes the sentence "*two victims are trapped*" as its syntactic complement. But this gives rise to a type-mismatch: The complement is of type t (28b), whereas the modal is seeking a propositional argument of type $\langle s,t \rangle$ (28c). In such cases, an additional mechanism kicks in, allowing the modal to combine with the intension of its complement. This is achieved by positing, in addition to extensional function application (29), a composition rule that introduces intensions when so required to avoid a crash of the semantic derivation (33). The derivation of truth-conditions of (32) then proceeds as in (34).

(32) [[might f_7] [$_{\text{TP}}$ [$_{\text{DP}}$ two victims] [$_{\text{VP}}$ trapped]]]

(33) **Intensional Functional Application (IFA):** If α is a branching node with daughters $\{\beta, \gamma\}$, then for any world w, if $[\![\beta]\!]^{w}$ is a function whose domain contains $[\lambda w'.\ [\![\gamma]\!]^{w'}]$, then $[\![\alpha]\!]^{w} = [\![\beta]\!]^{w}(\lambda w'.\ [\![\gamma]\!]^{w'})$.

(34) $[\![(32)]\!]^{w,g} = [[\![\text{might } f_7]\!]^{w,g}(\lambda w'.[\![[_{\text{TP}}\ [_{\text{DP}}\ \text{two victims}]\ [_{\text{VP}}\ \text{trapped}]]]\!]^{w',g})]$
$= \exists w' \in g(7)(w) : \exists x : |x| = 2 \wedge \text{victim}(w',x) \wedge \text{trapped}(w',x)$
(substitution from (28b-28c))

IFA—the mechanism that introduces intensions into the derivation (33)—is only triggered by the need to provide an intensional operator (e,g., a modal) with the input it is looking for (e.g., a proposition). In other words, intensions are only introduced into the derivation when we are interpreting a node immediately dominating an intensional operator. As a result, *all* and *only* elements in the syntactic scope of such an operator will receive opaque evaluation relative to that operator. Therefore, for the LF in (32), in which the DP "*two victims*" is positioned below the modal "*might*", the system only derives truth-conditions that correspond to the *de dicto* reading of (15). To derive a *de re* reading in which the restrictor "*victims*" is evaluated transparently, we need another LF, in which that DP moves to a position above the intensional environment created by the modal, provided in (35). Together with the composition rules that allow us to interpret such movement structures (30-31), and the generalized quantifier semantics for the DP in (36), we end up with the truth-conditions in (37) for (35).[14]

14 Derivation omitted for conciseness.

(35) $[[_{\text{DP}}$ two victims$]$ $[\lambda_{15}$ $[$[might f_7] $[_{\text{TP}}$ $[x_{15}]$ $[_{\text{VP}}$ trapped$]]]]]$

(36) $[\![\text{two victims}]\!]^{w,g} = \lambda f.\ \exists x : |x| = 2 \wedge \text{victims}(w,x) \wedge f(x)$

(37) $[\![(35)]\!]^{w,g} = \exists x : |x| = 2 \wedge \text{victims}(w,x) \wedge \exists w' \in g(7)(w) : \text{trapped}(w',x)$

Recall that the main shortcoming of STI, as discussed abstractly in Section 4, is its biconditional dependence between quantificational scope and intensional status which undergenerates Fodor's *third reading*, which pairs narrow quantificational scope with transparent evaluation. This should now be immediately observable: Since anything below an intensional operator is interpreted opaquely relative to the operator and anything above it is interpreted transparently, movement of a restrictor is required to achieve transparent evaluation. But, on the common assumption that movement of the restrictor necessitates movement of the determiner that it restricts, in the resulting system a determiner will necessarily take wide scope whenever its restrictor is transparent.

The undergeneration problem for STI is ameliorated in Keshet 2008, 2010 by a minimal modification which licenses transparent evaluation of elements below an intensional operator, but does not do away with the core principle of STI that syntactic position determines intensional status, and does not enrich the system with world pronouns of type s. Such a system, which Keshet calls *Split Intensionality*, enriches our vocabulary with an operator, $\hat{}$, defined syncategorematically as in (38), which returns the intension of its input. For convenience, I will sometimes say that $\hat{}$ *intensionalizes* its input.

(38) For any expression α and world w, $[\![\hat{}\alpha]\!]^w = \lambda w'.\ [\![\alpha]\!]^{w'}$

This operator replaces IFA as the mechanism responsible for introducing intensions, and thus equips us with a way of introducing intensions that is not parasitic on the immediate presence of an intension-seeking operator. In other words, it lets us form an intension-denoting node by inserting $\hat{}$ *anywhere* in the structure below an intensional operator, while still preserving a region in the structure—namely, the region between $\hat{}$ and the nearest intensional operator above it—where elements are interpreted transparently.

For a less abstract illustration, let us derive in Keshet's system the third reading of, say Fodor's (12), repeated in (39), which contains an attitude predicate as its intensional operator.[15] I assign the LF in (40) to the sentence, in which the DP "*an inexpensive coat*" raises to a position between the attitude "*want*" and $\hat{}$. This

15 I must briefly depart here from our example in (15), because that sentence involves two existential quantifiers; the modal and the plural indefinite. But these quantifiers are scopally commutative, and so the third reading of the sentence is semantically indistinguishable from its *de re* reading in (37). I therefore take this detour to illustrate how the third reading is derived in Keshet's system and will return to our target example below.

derives, for the sister node of the raised DP, the denotation in (41a), which is a function of type $\langle e, st\rangle$. Given that the DP itself denotes a generalized quantifier of type $\langle et, t\rangle$, the two cannot combine via FA (29). Rather, an additional composition rule is posited, which lets us derive the proposition in (41b). I will use a simplified version of such a rule, reproduced in (42) from Elliott 2023, which lets us "pass up" the world argument of the function in (41a) before composing it with the generalized quantifier. The proposition in (41b) in turn constitutes the input to the attitude verb, which is thus able to scope above the indefinite DP even though that DP's restrictor is evaluated transparently, resulting in the truth-conditions in (41c).

(39) Mary wants to buy an inexpensive coat.

(40) [Mary [wants [[an inexpensive coat] [7 [^ [PRO$_{\text{Mary}}$ to buy x_7]]]]]]

(41) a. $[\![[7\ [\hat{}\ [\text{PRO}_{\text{Mary}} \text{ to buy } x_7]]]]\!]^w = \lambda x \lambda w'$. Mary buys x in w'

b. $[\![[[\text{an inexpensive coat}]\ [7\ [\hat{}\ [\text{PRO}_{\text{Mary}} \text{ to buy } x_7]]]]]\!]^w =$
$\lambda w'.\ \exists x : x$ is an inexpensive coat in $w \wedge$ Mary buys x in w'

c. $[\![(40)]\!]^w = \forall w' \in \text{DESIRE}_{Mary,w}$:
$\exists x : x$ is an inexpensive coat in $w \wedge$ Mary buys x in w'

(42) **Saturation (SA):** If, for any world w, α is a branching node with daughters $\{\beta, \gamma\}$, with $[\![\beta]\!]^w$ a function of type $\langle et, t\rangle$ and $[\![\gamma]\!]^w$ a function of type $\langle e, st\rangle$, $[\![\alpha]\!]^w = \lambda w'.\ [\![\beta]\!]^w(\lambda x.\ [\![\gamma]\!]^w(x)(w'))$.

Now let us return to our target sentence in (15). Its *de dicto* reading, within Keshet's system, is derived almost exactly in the same way as it is derived in STI, illustrated in (32-34). The only difference, of course, is that in the modified system there is no IFA rule, so intensions are introduced by inserting the ^ operator at LF, as in (43).

(43) [[might f_7] [^ [$_{\text{TP}}$ [$_{\text{DP}}$ two victims] [$_{\text{VP}}$ trapped]]]]

As for R4, there are several technically interesting ways to derive this unattested reading for (15) in Keshet's system, of which I present next the one whose presentation seems the most simple to me. All of the possible derivations require a slightly more significant departure from the *de dicto* LF in (43), relative to the what we have seen above in our discussion of the world variable framework. What is important, though, is that the reading is still derivable from a structure in which the DP "*two victims*" does not outscope the existential modal, by inserting cumulation and ^ operators in the right positions, and slightly expanding the set of composition rules that let us "pass up" world arguments, which Keshet uses as well.

One LF that would derive R4 for (15) in this system is provided in Figure 5. In this structure, we use the intensionalizing operator ^ to create nodes that denote

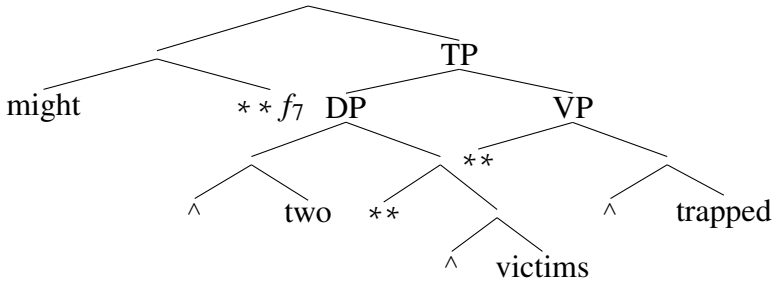


Figure 5 LF that derives R4 for (15), in Keshet's scope system

intensions, which are then supplied as input to a cumulation operator. More specifically, ^ attaches to the lexical predicates "*victims*" and "*trapped*", which in any scope system denote functions of type $\langle e,t\rangle$ (28a), resulting in intensions of type $\langle s,et\rangle$, as spelled out in (44).

(44) a. $[\![\text{\^{}victims}]\!]^w = \lambda w'\lambda x.\ \text{victims}(w',x)$

b. $[\![\text{\^{}trapped}]\!]^w = \lambda w'\lambda x.\ \text{trapped}(w',x)$

Of course, attaching ^ to a lexical item in the split intensionality system results in the same kind of intension that constitutes the denotation of that same lexical item in the world variable system; e.g., $[\![\text{\^{}victims}]\!]^w$ in (44a) is just what we have in (22a) for $[\![\text{victims}]\!]$. And we have already seen, in our discussion of the world variable system, the results of applying $**$ to such intensions. These are repeated in (45a-45b) below, now as the denotation of "double-starred" intensionalized predicates.

(45) a. $[\![**\text{\^{}victim}]\!]^w = \lambda w'\lambda x.\ [\forall y \leqslant x : \exists w'' \leqslant w' : \text{victim}(w'',y)] \wedge [\forall w'' \leqslant w' : \exists y \leqslant x : \text{victim}(w'',y)]$ (≈ (25a))

b. $[\![**\text{\^{}trapped}]\!]^w = \lambda w'\lambda x.\ [\forall y \leqslant x : \exists w'' \leqslant w' : \text{trapped}(w'',y)] \wedge [\forall w'' \leqslant w' : \exists y \leqslant x : \text{trapped}(w'',y)]$ (≈ (25b))

With these denotations for the pluralized intensionalized predicates, there are two potential type-mismatches in Figure 5. We run into the first when attempting to combine the intension of type $\langle s,et\rangle$ in (45a), within the DP in Figure 5, with the determiner "*two*". As before, we still treat the determiner as just a generalized quantifier of type $\langle et,\langle et,t\rangle\rangle$ (46). However, we also attach to it an intensionalizing operator, which vacuously binds a world argument (47), turning it into a function of type $\langle s,\langle et,\langle et,t\rangle\rangle\rangle$. This step does not, by itself, allow us to avoid type-mismatch, as intensions of type $\langle s,et\rangle$ are not in the domain of functions of type $\langle s,\langle et,\langle et,t\rangle\rangle\rangle$. However, it does allow us to posit a composition principle, which effectively lets us "ignore" the world argument, by passing it up to be saturated later in the derivation.

This is spelled out in (48), which lets us derive (49) as the denotation of the DP in Figure 5.[16]

(46) $[\![\text{two}]\!]^w = \lambda P_{\langle e,t\rangle}\lambda Q_{\langle e,t\rangle}.\ \exists x : |x| = 2 \wedge P(x) \wedge Q(x)$

(47) $[\![\hat{}\text{two}]\!]^w = \lambda w'.\ [\![\text{two}]\!]^{w'} = \lambda w'\lambda P_{\langle e,t\rangle}\lambda Q_{\langle e,t\rangle}.\ \exists x : |x| = 2 \wedge P(x) \wedge Q(x)$

(48) **World Percolation (WP):** If, for any world w and any types σ, τ, α is a branching node with daughters $\{\beta, \gamma\}$, with $[\![\beta]\!]^w$ a function of type $\langle s, \langle \sigma, \tau\rangle\rangle$ and $[\![\gamma]\!]^w$ a function of type $\langle s, \sigma\rangle$, $[\![\alpha]\!]^w = \lambda w'.\ [\![\beta]\!]^w(w')([\![\gamma]\!]^w(w'))$.

(49) $[\![\text{DP}]\!]^w = \lambda w'\lambda P_{\langle e,t\rangle}.\ \exists x : |x| = 2 \wedge P(x) \wedge [\forall y \leqslant x : \exists w'' \leqslant w' : \text{victim}(w'', y)] \wedge [\forall w'' \leqslant w' : \exists y \leqslant x : \text{victim}(w'', y)]$

We can now appeal to WP to avoid a second potential type-mismatch, when attempting to combine the DP in (49), of type $\langle s, \langle et, t\rangle\rangle$, with the VP, which is again a intension of type $\langle s, et\rangle$. WP lets us derive the denotation in (50) for the TP in Figure 5, again percolating up the world argument, and run-of-the-mill function application then derives (51) as the result of combining that TP with the modal.

(50) $[\![\text{TP}]\!]^{w,g} = \lambda w'.\ \exists x : |x| = 2 \wedge [\forall y \leqslant x : \exists w'' \leqslant w' : \text{victim}(w'', y)] \wedge [\forall w'' \leqslant w' : \exists y \leqslant x : \text{victim}(w'', y)] \wedge [\forall y \leqslant x : \exists w'' \leqslant w' : \text{trapped}(w'', y)] \wedge [\forall w'' \leqslant w' : \exists y \leqslant x : \text{trapped}(w'', y)]$

(51) $[\![\text{Figure 5}]\!]^{w,g} = [\![\text{might } f_7]\!]^{w,g}([\![(50)]\!]^{w,g}) = 1$ *iff* $\exists w' \in [\![**]\!]g(7)(w)$: $\exists x$: $|x| = 2 \wedge [\forall y \leqslant x : \exists w'' \leqslant w' : \text{victim}(w'', y)] \wedge [\forall w'' \leqslant w' : \exists y \leqslant x : \text{victim}(w'', y)] \wedge [\forall y \leqslant x : \exists w'' \leqslant w' : \text{trapped}(w'', y)] \wedge [\forall w'' \leqslant w' : \exists y \leqslant x : \text{trapped}(w'', y)]$

As already illustrated in our discussion of the world variable system above, the truth conditions in (51) are met in the collapse scenario (16), which verifies exclusively the R4 of (15). Therefore, if the structure in Figure 5 is available as a LF for (15), and the additional composition principle we posit here—namely, WP (48)—is legitimate, then the split intensionality system overgenerates unattested instances of R4, just like the world variable system.[17]

16 While I define WP in (48) in a way that is specifically tailored to our needs in this section, it is actually an instantiation of a general operation in combinatory logic (the S-combinator; cf. Seldin 2006), that has been appealed to in work within combinatorial grammar (cf. Steedman 1987, 1988; Szabolcsi 1989). Note further that the same instantiation is appealed to in, e.g., Schmitt's (2020) work, as a component in the derivation of certain readings of attitude constructions.

17 A potential response on behalf of Split Intensionality might target our distribution of intensionalizing operators in Figure 5. For instance, $\hat{}$ is applied vacuously to the determiner "*two*", and one could bar this kind of move. Alternatively, one may argue that the $\hat{}$ operators we attach to predicates of

5.3 Taking stock

We have seen how two different approaches to the representation of intensionality overgenerate the unattested R4 once augmented with a mechanism that derives cumulative inferences. While the derivations of R4 differ in their technicalities under each approach, they share a simple common thread. In both cases, what enabled the generation of R4 is our ability to close under sum-formation two kinds of intensions:

(i) relations between worlds and individuals contributed by nouns or verbs like "*victim*" or "*trapped*", and

(ii) functions from worlds to truth values, serving as restrictors of modals like "*might*".

We have assumed, as is standard, that in both systems intensions of the type described in (ii) are simply supplied contextually by syntactic variables that range over functions of type $\langle s,t\rangle$. As for the intensions described in (i), these are by definition the denotations of nouns and intransitive verbs in the variable framework, and are derived via an intensionalizing operator at LF under split intensionality.

Our findings provide new support for Schmitt's (2023) conclusion that there is a ban on cumulative readings with worlds, as such a ban would be needed to block the derivation of R4, under either theory of intensionality. As noted earlier, unlike Schmitt's original argument from the lack of cumulative readings of conditionals and other particular modal constructions, the argument above cannot be bypassed by stipulating that particular lexical items behave in a particular way. Rather, a more general stipulation would be needed, prohibiting (at the very least) any intension of the kind in either (i) or (ii) from undergoing closure under sum-formation.

One question that immediately arises, of course, is whether the Scope Theory

type $\langle e,t\rangle$ are illicit. These operators create two nodes that denote relations of type $\langle s,et\rangle$, which are then pluralized to achieve R4. However, all that Keshet really needs to derive Fodor's (1970) third reading, is to place $^\wedge$ in a position whose sister is a node of type t as in (40), creating a proposition-denoting node. And while no deep principle in the system restricts $^\wedge$ to such positions, perhaps our derivation can be blocked by positing such a restriction. There may even be independent motivation for doing so: Larson (2002) and den Dikken, Larson & Ludlow (2018) argue, from the behavior of so-called intensional transitive verbs, that intensionality is grammatically conditioned in that it is introduced only at the level of clausal complements, suggesting that only syntactic clauses may be intensionalized (I thank REMOVED for pointing out this possible explanation to me). I will not engage with this point here besides, first, noting that it is quite controversial (for instance, Forbes (2000, 2006); Zimmermann (2006) present well-known alternative views; see also Schwarz 2021 for an overview of the debate and further references). Second, note further that even if this turns out to be a viable response to the challenge of R4, this response requires one be committed to a scope system as the correct framework with which to account for the world-dependence of utterances. This, in itself, is a notable result.

of Intensionality (STI) can avoid the overgeneration problem we observe here. After all, the main difference between STI and the two theories we have considered, which are themselves quite different from each other, is that under STI no LF nodes denote the intensions of the type described in (i). Rather, lexical items like "*victim*" or "*trapped*" denote extensions, and are intensionalized only in the semantics by the IFA composition rule that effectively functions as a "repair strategy", kicking in when required to avoid type-mismatch. Given that closing the intensions of the kind described in (i) under sum-formation is a necessary component in our derivations above, we should naturally wonder whether R4 may be blocked simply in virtue of the architecture of STI.

Next, I will show that, alas, STI does not block R4 *just* in virtue of its architecture. However, in the more speculative part of the paper to follow, I will spell out a possible way of blocking the unattested reading under STI by appealing to more general principles than those that must be appealed to in order to block the reading in the other frameworks we have considered. I will end the paper by elaborating on why this approach is promising, but also by listing some of the serious issues that future research must resolve for it to work.

5.4 The Scope Theory of Intensionality

What would be required to close a relation between worlds and individuals under sum-formation in a framework that does not make available LF nodes denoting such relations? As far as I can tell, there are two options. First, if the operators of the kind we have been using to derive cumulative inferences (i.e., $*$, $**$ etc.) can themselves have an intensional type, they would trigger the application of the intensionalizing composition rule IFA when combining with their argument, resulting in the sum-closure of the *intension* of their argument. Alternatively, suppose we assume the view termed *lexical cumulativity* in the literature, according to which predicates are "born" as closed under sum-formation. Lexical cumulativity itself might allow cumulation not only over subcategorized syntactic arguments, but over the world parameter as well. Let us review both of these possibilities.

Suppose we posit a cumulation operator that is itself a function seeking an intensional input; e.g., $**_{\text{INT}}$ in (52a). Merging this operator with the predicate "*victims*" in (52b) would trigger IFA (33) and derive the same function we derive with $[\![**\hat{}\text{victims}]\!]^w$ in (45a) within the system of split intensionality, or with $[\![**\text{victims}]\!]$ in (25a) within the world variable framework.

(52) a. $[\![**_{\text{INT}}]\!]^w = \lambda\mathcal{R}_{\langle s,\langle e,t\rangle\rangle}\lambda w'\lambda x.\ \exists\mathcal{R}'_{\langle s,\langle e,t\rangle\rangle} : [\forall w'',x' : \mathcal{R}'(w'',x') \rightarrow \mathcal{R}(w'',x')] \wedge$
$w' = \bigoplus\{w \mid \exists x : \mathcal{R}'(w,x)\} \ \wedge\ x = \bigoplus\{x \mid \exists w : \mathcal{R}'(w,x)\}$

b. $[\![\text{victims}]\!]^{w,g} = \lambda x.\ \text{victim}(w,x)$

c. $[\![**_{\text{INT}} \text{ victims}]\!]^w =_{\text{(by IFA)}} [\![**_{\text{INT}}]\!]^w (\lambda w'.\ [\![\text{victims}]\!]^{w'})$

Now, some non-trivial technical maneuvers—of the kind we had to resort to in deriving R4 in the framework of split intensionality—would be required to combine the resulting predicate, of type $\langle s, \langle e,t \rangle \rangle$ any further with a determiner like "*two*". But even with a plausible way to avoid type-mismatch, there is still a natural way of thinking about cumulation operators that would prevent us from positing intensional versions of these operators under STI.

Cumulation operators, like $*$ or $**$, must be polymorphic; i.e., must be able to apply to elements of different types (cf. Krifka 1986). However, how should we model their polymorphic nature? There are two possibilities. On the one hand, we may simply assume that there are multiple entries for $*$, $**$ etc. in the lexicon, such that for any conjoinable type σ (i.e., any type that ends with t), there is a $*$ or $**$ of type $\langle \sigma, \sigma \rangle$. This would mean that we have $**$ of, say, type $\langle \langle e,et \rangle, \langle e,et \rangle \rangle$, but also of type $\langle \langle s,et \rangle, \langle s,et \rangle \rangle$ as in (52a). On the other hand, we may assume that for an operator, being polymorphic means not having the lexicon fully determine its type; rather a polymorphic operator's type is only determined by the elements with which it combines in the syntax. This view is implicit in our entries for $*$ and $**$ in (19), repeated below, where instead of type labels, we use variables σ, τ over types.

(53) a. $[\![*]\!] = \lambda \mathcal{F}_{\langle \sigma,t \rangle} \lambda x_\sigma .\ \exists \mathcal{F}'_{\langle \sigma,t \rangle} : [\forall x' : \mathcal{F}'(x') \rightarrow \mathcal{F}(x')] \wedge x = \bigoplus \{x \,|\, \mathcal{F}'(x)\}$

b. $[\![**]\!] = \lambda \mathcal{R}_{\langle \sigma, \langle \tau,t \rangle \rangle} \lambda x_\sigma \lambda y_\tau .\ \exists \mathcal{R}'_{\langle \sigma, \langle \tau,t \rangle \rangle} : [\forall x', y' : \mathcal{R}'(x',y') \rightarrow \mathcal{R}(x',y')] \wedge$
$x = \bigoplus \{x \mid \exists y : \mathcal{R}'(x,y)\} \wedge y = \bigoplus \{y \mid \exists x : \mathcal{R}'(x,y)\}$

On this view, there can be an operator $**$ of type $\langle \langle s,et \rangle, \langle s,et \rangle \rangle$ like the one in (52a) only if there are nodes whose denotations are of type $\langle s,et \rangle$ in the syntax. And since STI does not let us construct such nodes to begin with, pairing this view of the polymorphic nature of cumulation operators with STI does not generate unattested cases of R4. Notably, unlike the stipulation that specifically prohibits intensions of type $\langle s,et \rangle$ from being closed under sum-formation, which would be required to block R4 as a cumulative inference under the world variable or split intensionality framework, under STI we can posit a general and natural constraint on polymorphic operations, from which the more specific stipulation immediately follows.[18]

18 The idea that the type of polymorphic operators is only determined once their argument position is saturated can be implemented by revisiting the idea that such elements are *operators* in the syntax to begin with. Instead, we can introduce the semantics of so-called "polymorphic operators" as composition rules. For instance, $**$ can be introduced with a rule defined as follows:

(i) $**$**-Rule:** If, for any world w and any types σ, τ, α is a non-branching node with daughter β, such that $[\![\beta]\!]^w$ is a function of type $\langle \sigma, \langle \tau,t \rangle \rangle$, one of the following holds:

- $[\![\alpha]\!]^w = [\![\beta]\!]^w$; or

Let us now turn to the more challenging case. Recall that we have chosen to derive cumulative inferences with operators like $*$ and $**$ because almost everyone in the literature seems to agree that we need a cumulation mechanism with, *at least*, the expressive power of the operator system to correctly capture the empirical landscape. The logic behind this choice, spelled out in Section 5.1, is that given the consensus that operators set the lower bound on required expressive power, if we can show that operators run into an overgeneration problem, that problem will apply to any more expressive system as well. We also mentioned another view, that of *lexical cumulativity*, according to which predicates are closed under sum-formation as part of their lexical meaning. That view, we noted, is subsumed by the operator approach, because any cumulative inference derived by assuming that the arguments of some predicate are closed under sum-formation in the lexicon, could also be derived by assuming closure under sum-formation occurs via an operator adjoining that predicate at LF.

However, once we introduce assumptions that effectively constrain the expressive power of the operator system, we should reexamine whether the resulting system is strictly more expressive than the lexical view. Indeed, complementing the operator approach with the lexical view might allow the former to make up for whatever expressive power it loses by pairing STI with the view of polymorphic operators described above (Viola Schmitt, p.c.). This is because, if closure under sum-formation is a lexical property of predicates, nothing in principle blocks it from targeting the world parameter of a predicate in addition to its subcategorized arguments. More concretely, the lexical entries for items like "*victim*" and "*trapped*" may be defined as in (54a-54b), and similarly the $\langle s,t\rangle$ restrictor of a modal like "*might*" might itself be "born" as closed under sum-formation.

(54) a. $[\![\text{victim}]\!]^w = \lambda x.\ [\forall y \leqslant x : \exists w' \leqslant w : \text{victim}(w',y)] \wedge [\forall w' \leqslant w : \exists y \leqslant x : \text{victim}(w',y)]$

b. $[\![\text{trapped}]\!]^w = \lambda x.\ [\forall y \leqslant x : \exists w' \leqslant w : \text{trapped}(w',y)] \wedge [\forall w' \leqslant w : \exists y \leqslant x : \text{trapped}(w',y)]$

This will suffice to derive R4 truth conditions for our target sentence in (15). Here is how: Assuming the structure from which we derived the *de dicto* reading of (15) under STI, repeated from (32) in (55a), and the entries in (54a-54b) for "*victims*" and "*trapped*", we derive (55b) as the denotation of TP. This is a truth value which combines with "*might*" via IFA. As long as the restrictor of "*might*", contributed by the free pronoun f_7 in (55a) contains all epistemically-accessible

- $[\![\alpha]\!]^w = \lambda x_\sigma \lambda y_\tau . \exists \mathcal{R}'_{\langle\sigma,\langle\tau,t\rangle\rangle} : [\forall x', y' : \mathcal{R}'(x',y') \rightarrow [\![\beta]\!]^w(x',y')] \wedge x = \bigoplus\{x \mid \exists y : \mathcal{R}'(x,y)\} \wedge y = \bigoplus\{y \mid \exists x : \mathcal{R}'(x,y)\}$

I thank REMOVED for discussing this point with me.

worlds and their pluralities, the sentence thus asserts that there is a world or plurality thereof of which the proposition $\lambda w.[\![\text{TP}]\!]^{w,g}$ holds, and this in turn is the case in the R4-scenario that falsifies (15).

(55) a. [[might f_7] [$_{\text{TP}}$ [$_{\text{DP}}$ two victims] [$_{\text{VP}}$ trapped]]]

b. $[\![\text{TP}]\!]^{w,g} = 1$ iff $\exists x : |x| = 2 \wedge [\forall y \leqslant x : \exists w' \leqslant w : \text{victim}(w', y)] \wedge [\forall w' \leqslant w : \exists y \leqslant x : \text{victim}(w', y)] \wedge [\forall y \leqslant x : \exists w' \leqslant w : \text{trapped}(w', y)] \wedge [\forall w' \leqslant w : \exists y \leqslant x : \text{trapped}(w', y)]$

So STI seems to require a stipulation to block the derivation of unattested instances of R4—say, that the world parameter is never cumulated over—that appears just as *ad hoc* as the one required under split intensionality or the world variable framework.[19]

This concludes the main argument of the paper, which shows that no matter which framework for representing intensionality we choose, we end up overgenerating unattested instances of R4 by cumulation with pluralities of worlds. Thus, to correctly capture what many subscribe to as the correct generalization regarding intensionality and scope, we must also accept the less common position that intensions cannot be closed under sum-formation. Next comes the more speculative part of the paper, which explores whether STI may be better positioned to block R4 by appealing to independently motivated constraints on cumulation.

6 The scope theory after all?

So far we have seen that under any of the three intensional systems introduced above, one can derive an unattested R4 for the sentence in (15). However, there is a crucial difference between the derivations of R4 with world variables (Section 5.2.1) or split intensionality (Section 5.2.2), and its derivation under STI (Section 5.4). In the latter case, given the absence of intension-denoting nodes at LF and the natural view of polymorphic operators discussed above, deriving R4 committed us to a particular view of cumulation; namely, to the view that predicate extensions are closed under sum-formation in the lexicon. Derivations with world variables and split intensionality required no such assumptions; while we used syntactic cumulation operators to derive R4 in these systems, this was a presentational choice, *not* a

19 Of course, a constraint blocking lexical cumulation over a world parameter or argument is required under *any* of the intensional systems we discuss: If lexical cumulativity lets us close under sum-formation intensions in the world-variable system, then regardless of how we manage to block cumulation operators from applying to these intensions, we will still derive R4. The same point applies, *mutatis mutandis*, to Keshet's (2008) system.

theoretical commitment. The derivation of R4 under STI would thus be blocked if we have reason to believe that lexical cumulativity is itself unavailable.

Incidentally, there is new work in the plurality literature which suggests exactly that. In particular, Wehbe (2023, 2026), building on Ionin & Matushansky (2002), recently argued that cumulation should be derived *only* via syntactic operations, never in the lexicon, and conjectures that this is due to a general constraint preventing certain meanings from being lexically encoded. Let us briefly present this argument, and then weigh some pros and cons of adopting its conclusions in an attempt to block unattested R4s under STI.

The argument is quite intricate, so I will accordingly only reproduce here its core observation. It involves the Lebanese Arabic *double subject construction*, exemplified in (56a), in which one thematic subject (Rasha and Hadi), is realized with two DPs. This construction is argued to differ structurally from run-of-the-mill sentences with a plural subject like (56b) only in the position of the conjoined DP subject; in (56a), that DP is in a position below the surface position of the verb, and non-standard movement of a single conjunct above the verb results in the observed word order, whereas in (56b) the conjoined DP itself moves to a pre-verbal position. It is then observed that the structural difference is correlated with a difference in meaning: While (56b) can be true if Rasha and Hadi danced either together (one with the other), or separately, (56a)—with the double subject—is only verified when the dancing was collective.

(56) a. **Rasha** raʔasit **hiyye w Hadi.**
Rasha danced.3SG.F **her and Hadi**
'Rasha and Hadi danced with each other.'
Simplified structure: [Rasha [$_{XP}$ λ_1 [danced [her$_1$ and Hadi]]]]

b. **Rasha w Hadi** raʔaso.
Rasha and Hadi danced.3PL
'Rasha and Hadi danced.'
Simplified structure: [[Rasha and Hadi] [$_{XP}$ λ_1 [danced t_1]]]

But if it is a lexical property of the verb "*dance*" that whenever it holds of Rasha and Hadi separately, it holds of their conjunction, the difference in meaning is not predicted: In that case, the node labeled XP in (56a) would denote a predicate true of any individual x if the sum $x \oplus$Hadi is in the denotation of the verb "*dance*". This predicate holds of Rasha as long as she and Hadi danced, separately *or* together, failing to derive the unique meaning of (56a).

Wehbe argues that these facts support the following claims. First, that sum-formation never takes place in the lexicon; rather, a plurality is in the denotation of a lexical predicate only if the predicate holds of the members of this plurality collectively (in the case of "*dance*", only if the members of the plurality danced together).

Second, that sum-formation is syntactically restricted to derived predicates; thus, in (56a-56b) only the nodes labeled XP can be the input of cumulation operators like $*$ and $**$, but not the lexical predicate "*dance*" itself.[20] Given these assumptions, the node labeled XP in (56a) denotes a predicate true of an individual x if x and Hadi danced collectively. In (56b), XP denotes a predicate true of an individual or plurality x if x danced, and since XP itself can be closed under sum-formation via $*$ it will hold of the plurality of Rasha and Hadi as long as both individuals danced, separately or together. Wehbe conjectures that the absence of lexical cumulativity may be due to an economy constraint on the lexicon, ruling out lexicalizing meanings that can be derived by the application of an independently needed operator. And given that a sum-closure operator is independently needed (Beck & Sauerland 2000), it must be solely responsible for closure under sum-formation.

If these conclusions are warranted, then while the principles of STI do not by themselves block the derivation of R4, they can do so together with the general constraint ruling out lexical cumulativity that Wehbe independently motivates. However, the logic implied by Wehbe's economy constraint, ruling out closure under sum-formation in the lexicon given that the same meanings can be derived via the application of $*$ or $**$, does not fit as neatly with the STI account of the absence of R4 as one would have hoped: Under STI, the meaning that is derived by cumulating over the world parameter in the lexicon is one that (given our assumption about polymorphic operators) *cannot* be derived by the application of a syntactic operator. So while we must accept the conclusion that lexical cumulativity is ruled out, we

20 One may legitimately protest here, wondering whether Wehbe's assumption that cumulation operators only apply to derived predicates does not by itself already rule out the derivation of the R4 of (15) under the world-variable system, which would render moot our entire discussion of STI. To derive R4 with world variables, we posited the structure in Figure 4, where $**$ applies directly to lexical predicates denoting intensions like "*victims*" and "*trapped*". However, we can introduce certain technically creative abstraction steps to the structure in Figure 4, which will effectively form derived predicates equivalent in meaning to the lexical predicates that, according to Wehbe, cannot be closed under sum-formation. For instance, suppose that instead of mapping the DP "*two victims*" to the structure in (i-a) as we do in Figure 4, we assign to it the structure in (i-b), where the determiner "*two*" had moved a short distance, leaving behind an individual-type trace, and the world variable in the restrictor then "tucked-in" between the determiner and its abstractor to create another abstraction node, over worlds. The resulting derived predicate is of type $\langle s, et \rangle$ and can be closed under sum-formation via $**$, as in (i-b). Wehbe argues at length that a technical maneuver of this sort is unavailable for the double-subject construction, by appealing to generalized economy considerations which in turn do not apply in our case. Thus, while this operation is non-standard, it is unclear to me what principled reasons might rule it out.

(i) a. λ_5 … $[_{\text{DP}}$ two $[**\text{victims } w_5]]$

b. λ_5 … $[_{\text{DP}}$ two $[w_5\ [* * \lambda_4\ \lambda_7\ [t_7\ [\text{victims } t_4]]]]]$

need to reevaluate the reason that Wehbe conjectures is behind this fact.

Here is another challenge facing this approach, which—we will show immediately—is actually related to the question of what constraint on the lexicon should do the job of ruling out lexical cumulativity. Under STI, how can we derive the true reading of the belief report in (6), repeated in (57b), in a scenario like (57a) in which neither John nor Bill believe the conjunction in the prejacent of the belief predicate, but each believes one of the conjuncts?

(57) a. **Context:** Sam owns a construction company. Bill and John are her clients. John believes Sam is from Texas, and has no opinion on whether she is rich, while Bill believes she is rich but does not know where she is from.

b. John and Bill believe that Sam is from Texas and that she is rich.
(Marty 2019)

If this reading is a cumulative one, as Marty (2019) and (Schmitt 2023) argue, it involves the belief relation holding cumulatively between a plurality of individuals, John and Bill, and the plurality of *propositions* denoted by the conjunction in its prejacent. But STI does not let us treat conjunctions as a pluralities of propositions to begin with, given that the prejacent denotes a simple truth value up until that point in which it combines with an intensional operator, which intensionalizes it via IFA.[21] And if we do find a way to treat a conjunctive prejacent as a plurality of propositions, what prevents us from formulating a lexical entry for any intensional operator that incorporates a cumulative relation between pluralities of worlds and propositional conjunctions into its meaning? For instance, consider the conditional operator "*if*", which, as Schmitt (2023) shows, cannot have a conjunction in its consequent and a plurality of worlds of which its antecedent holds partake in a cumulative *hold-of* relation (cf. (8-9)). A constraint ruling out lexical cumulativity does not rule out an entry like (58) for "*if*", where closure under sum-formation of intensions is introduced within the entry itself, which has a cumulative *hold-of* relation baked into it.

(58) a. $[\![\textit{if}]\!]^w = \lambda p, q \in \mathcal{D}_{\langle s,t\rangle}.1$ *iff* $\forall q' \leqslant q : \exists w' \in \text{MAX}([\![*]\!]\text{CLOSEST}(w)(p)) : q'(w') \wedge$
$\forall w' \leqslant \text{MAX}([\![*]\!]\text{CLOSEST}(w)(p)) : \exists q' \in q : q'(w')$

Definitions:

i. CLOSEST is a function $f : \mathcal{D}_s \times \mathcal{D}_{\langle s,t\rangle} \to \mathcal{D}_t$, s.t. $\forall w, p : \text{CLOSEST}(w)(p) =$ $\lambda w'.\ p(w') \ \wedge \ \forall w'' : p(w'') \to w'$ is at least as similar to w as w''

21 We could, perhaps, deny that the reading of this sentence that is true in the given scenario really is a cumulative one—opting, say, for Pasternak's (2018) account in terms of "joint belief" discussed in Section 2—but as I note there, in that case we would face the task of explaining why the reading seems to show hallmarks of cumulativity.

ii. MAX is a function $f : \mathcal{D}_{\langle\sigma,t\rangle} \to \mathcal{D}_{\sigma}$ (where σ is any type), s.t. $\forall X \in \mathcal{D}_{\langle\sigma,t\rangle} : \text{MAX}(X)$ is defined only if $\exists x \in X : \forall x' \in X : x' \leqslant x$ and if defined, $\text{MAX}(X) = \iota x \in X : \forall x' \in X : x' \leqslant x$ (i.e., the maximal element in X)

Or in words, $\text{MAX}([\![*]\!](\text{CLOSEST}(w)(p)))$ is the plurality of p-worlds most similar to w.

b. $[\![[\mathit{if}\ \phi]\ [\psi]]\!]^w =_{\text{by IFA}} [[\![\mathit{if}]\!]^w(\lambda w'.\ [\![\phi]\!]^{w'})](\lambda w'.\ [\![\psi]\!]^{w'}\ )$

These issues together suggest that a constraint blocking lexical cumulativity must not be contingent on the presence or absence of any alternative derivation, but rather, must prevent certain logical operations—crucially including closure under sum-formation—from *ever* being carried out in the lexicon or within the lexical entries assigned to object language expressions, full stop. And formulating such a constraint is far from trivial. These challenges led Schmitt (2023) to reject an STI-account of the ban on world-pluralities. Now, it might very well be that Wehbe's (2026) argument should make us reevaluate this rejection—given that it motivates a constraint that would be subsumed by the one we would need to block cumulative readings with worlds under STI—but obviously this must be more carefully examined; an endeavor I leave for future research.

Finally, probably the most difficult challenge hovering over an STI-account of this sort comes from the fact that the step of adopting STI itself may be a stipulation; simply calling something a parameter, rather than a variable, does not *explain* what the nature of this object is and how it differs from a variable. Therefore, even if we manage to derive the absence of R4, and of cumulative readings with worlds more generally, from STI in conjunction with independently motivated constraints on cumulation, the adoption of STI itself constitutes an ad hoc assumption that is unique to worlds. Until we have something explanatory to say about the distinction between variables and parameters, using STI to rule out R4 remains, in a sense, a technical exercise.[22]

But there is also a good reason to nevertheless think this approach is worth pursuing. In particular, recent literature suggests that deserting STI in the first place might not have been so well motivated. Recall that the main reason for adopting the more expressive systems considered here was that, given STI's biconditional dependence between the scope of a DP and the intensional status of its restrictor, it undergenerates Fodor's (1970) *third* reading, which involves transparent evaluation of restrictors of narrow scope DPs.

Interestingly, the literature on intensionality has converged recently on the position that there are independent reasons to posit an additional interpretative mechanism for intensional sentences; one that would derive the third reading and can

22 I thank Viola Schmitt for discussing this point.

supplement STI to avoid undergeneration. Kaufmann (in Schwager 2009) influentially argues that not all instances of the third reading reduce to shifting the world index of a DP in the scope of an intensional operator, contra the way we have been thinking about that reading so far, and the way Keshet (2008) and Percus (2000) think about it when proposing their alternatives to STI.[23] This means that *all* of the theories observed so far undergenerate at least some instances of the third reading.

Kaufmann's example is (59b), which seems true in the context of (59a), which falsifies its *de re* reading, as there is no specific building Mary wants to buy, and its *de dicto* reading, as Mary (by assumption) does not know how many floors the Burj Dubai has. Thus, the DP "*a building with 192 floors*" must scope below the attitude verb "*want*" but its NP restrictor must not be evaluated opaquely relative to that verb. This leaves us with the *third* reading. But if that reading simply pairs narrow scope for a DP with opaque evaluation of its restrictor, as formalized in (59c), (59b) should attribute to Mary inconsistent desires: the truth conditions in (59c) hold only when Mary desires the contradiction $\exists x \in \varnothing$ to hold, as there is no building with 192 floors in the evaluation world.

(59) a. Context: Mary is looking at the Burj Dubai. Not knowing that it has 191 floors and is the world's highest building, she thinks, 'I want to buy a building that's a floor higher!'

b. Mary wants to buy a building with 192 floors.

c. $[\![(59b)]\!]^w = 1$ *iff* $\forall w' \in \text{DESIRE}_{Mary,w}[\exists x\text{: } \textit{has-192-floors}_w(x) \wedge \textit{buy}_{w'}(\textit{Mary},x)]$

d. Mary wants to buy a building that is one floor higher than the Burj Dubai.

The fact of the matter is, however, that (59b) does not seem to convey that Mary has inconsistent desires. Rather, it seems to convey something along the lines of what (the *de dicto* reading of) (59d) does, which cannot be captured under the approach to the third reading taken for granted so far.

Kaufmann derives the correct reading of (59b), i.e., the *de dicto* reading of (59d), by positing a mechanism for property substitution that effectively replaces "*building with 192 floors*" in (59b) with "*building that is one floor higher than the Burj Dubai*", and evaluates the sentence *de dicto* after replacement has taken place. A flurry of recent work has attempted to characterize the replacement mechanism required for these proposals to work.[24] What matters for us is just that, whatever it is, this mechanism is able to supplement STI in a way that eliminates the challenge presumably posited by the third reading. And there seems to be a consensus that,

23 In fact, Kaufmann's position is that *none* of the instances of the third reading involve narrow scope and transparent evaluation.

24 See, in particular, Sudo 2014; Baron 2015; Percus 2020; Blumberg & Lederman 2021; Benbaji-Elhadad 2023, 2026a; Mayr & Schmitt 2024, 2026

in fact, once we have in place a substitution mechanism, we can—and perhaps should—use it beyond (59b) to derive all instances of the third reading. Now, since no theory we have examined so far is able to account for (59b), they all need to posit this additional mechanism, and if the assumptions regarding cumulativity that STI would needs to correctly capture the absence of R4 turn out to be well founded, alternative intensional systems will themselves turn out to be inferior to STI from the perspective of the fourth reading.

Note, further, that replacement is needed in a wider range of examples than just those involving the third reading. In particular, Fauconnier (1984) observed that we can truthfully ascribe to individual x belief in some proposition p even if x's belief state does not entail p, as long as p is entailed by x's beliefs taken together with some of the reporter's presuppositions. In (60), due to Percus (2020), for instance, John has no beliefs about Mary's whereabouts. Still, (60a) seems true, which Percus attributes to the fact that Mary having arrived is entailed by Sue's knowledge that Mary is on flight AF62 together with John's belief that AF62 has arrived. Percus conjectures that replacement of the sort needed for the third reading may be active here, letting us interpret the attitude report in (60a) as if it conveys what the *de dicto* reading of (60b) does, as long as certain conditions are met.[25]

(60) Context: Mary is on flight AF62. Sue and Carol want to know if she arrived. Sue asks John, who works at the airport but does not know Mary is traveling, whether flight AF62 landed. He says he believes it has. Sue reports to Carol:

a. John thinks that Mary has arrived.

b. John thinks that flight AF62 has arrived.

Now, if a replacement mechanism that is powerful enough to deliver the right reading for (60a) is assumed to supplement STI, then all potential undergeneration challenges that STI has been argued to face require a reevaluation. And if a considerable amount of them can be derived via replacement, then they do not constitute a challenge for STI in the first place. As this section illustrates, there is much more work to be done to turn the main observation in this paper into an argument *for* STI. What I attempted to do in this speculative discussion is to make the best case for why this work is worth the effort, by highlighting recent findings in both the cumulativity and intensionality literature that might be taken to motivate the ingredients of an STI-account of the absence of R4.

25 The phenomenon that Fauconnier (1984) identified has been dubbed *revisionist reporting* by Blumberg & Lederman, and has been shown to extends beyond belief; see Pryor 2004; Recanati 2012; Blumberg & Lederman 2021 a.o.

7 Conclusion

In this paper, I showed that any theory of intensionality that allows lexical items or nodes dominating them at LF to denote intensions of type $\langle s, et \rangle$ overgenerates an unattested R4 of modal sentences like (61), repeated from (15), once it is paired with independently motivated mechanisms that derive cumulative inferences via LF operators. This is because these mechanisms can close intensions under sum-formation resulting in a reading that is true in exclusively-R4-verifying scenarios. I further show that even a theory that does *not* posit intension-denoting nodes at LF—namely, the Scope Theory of Intensionality (STI)—can in principle generate the unattested reading if we assume that cumulation can take place in the lexicon, but point to some general considerations that may block the lexical approach to cumulation.

(61) Two victims might be trapped under the rubble.

This result generalizes Schmitt's (2023) recent finding, that certain linguistic expressions previously argued to denote pluralities of worlds do not seem to give rise to cumulative inferences. It lends further support to the claim that Schmitt defends, that we simply should not be able to form world pluralities in our semantic derivations—neither in the lexicon nor in the syntax—by showing that this new claim must hold for our theories of intensionality to be able to derive a generalization long treated by many as a virtual truism.

Since the inception of possible worlds semantics for natural language, there has been a steady stream of arguments against its representation of propositions, information states, and content more generally as sets of worlds. These arguments all suggest that possible-worldly representations are too course-grained; they do not furnish us with the tools to distinguish between sentences expressing equivalent propositions, which—as a matter of fact—do pattern differently in certain linguistic contexts. Many have, consequently, gravitated towards a view of content whose primitive elements are not worlds, but rather finer-grained entities like situations, states of affairs, or truthmakers (for some instructive examples, see, *amo*, Kratzer 2007/2023; Yablo 2014; Fine 2017; Moltmann 2020; Champollion 2025). Semantic formalisms that make use of such finer-grained representations are all more expressive than possible worlds semantics, given their ability to draw distinctions that are obscured by possible-worldly representations.

This paper swims against that popular current: The argument it presents illustrates that possible worlds semantics is already itself *too expressive*, giving rise to unattested readings simply in virtue of the fact that it construes the denotations of certain lexical items as functions on worlds. As far as I can tell, the argument can thus be directed, *mutatis mutandis*, at any theory in which functions on any finer-grained entities constitute the denotations of any linguistic material. This tension,

between more expressive theories—and the data that motivates them—and the observation here, is an issue for future work on intensionality to resolve.

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